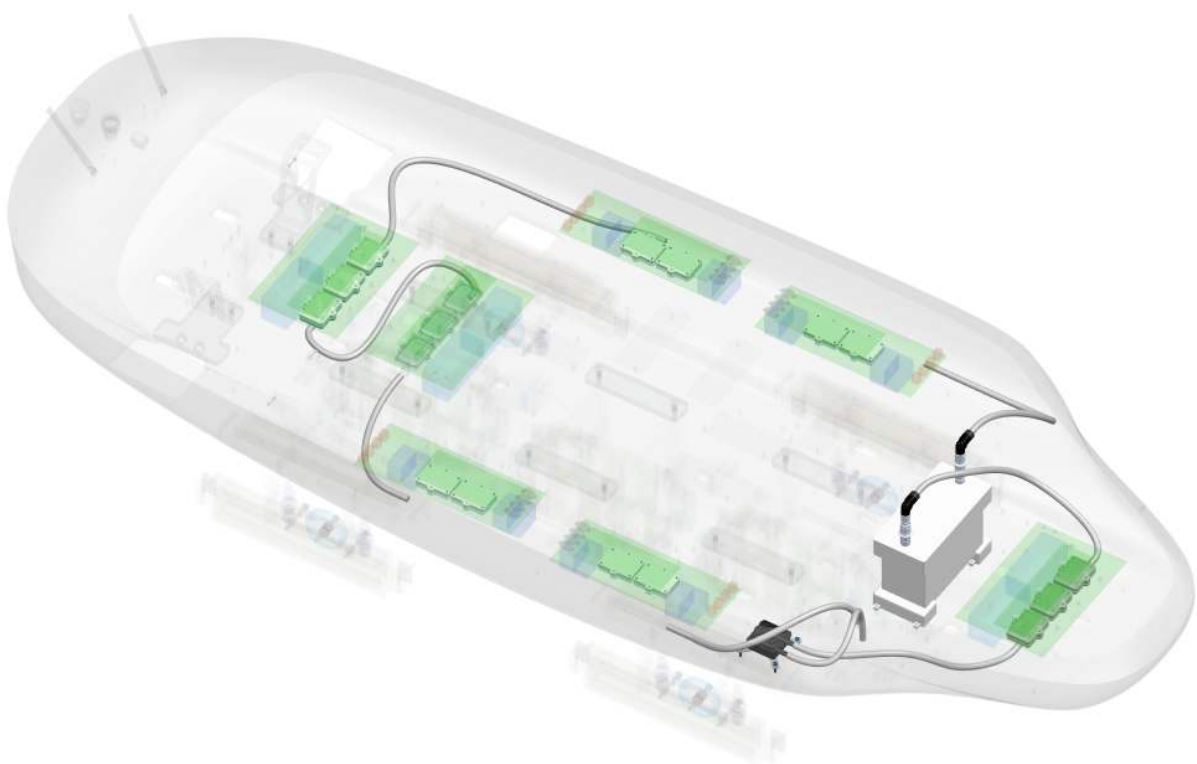


Design, Simulation and Manufacturing of a Phase Change Cooling Module for Low-Pressure Applications

Tim Aebersold
taebersold@student.ethz.ch

Bachelor's Thesis

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Supervisor
Professor

Dr. Jan Seiler, jaseiler@ethz.ch
Prof. Dr. André Bardow, abardow@ethz.ch

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Abstract

Thermal management in Hyperloop systems is a significant challenge due to low pressure environments which impede convective heat transfer. This thesis explores suitable concepts for thermal management, with a focus on latent heat storage using phase change materials (PCM).

A fully functional PCM cooling system with a weight of 6.5 kg and peak power capacity of 1750 W (400 W continuous) was designed and manufactured. The system is designed for a small-scale Hyperloop test vehicle for use in student competitions. It employs a paraffin PCM in a selective laser sintered (SLS) enclosure. Simulations were used to optimize and characterize the system's performance. The system was commissioned successfully. The potential of alternative approaches for thermal management, including water tanks, radiation and convection was also discussed.



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List of Acronyms

CF	Carbon Fiber
CFD	Computational Fluid Dynamics
CFRP	Carbon Fiber Reinforced Polymer
CNC	Computerized Numerical Control
EDS	Electrodynamic Suspension
EG	Expanded Graphite
EMS	Electromagnetic Suspension
EPDM	Ethylene Propylene Diene Monomer
FDM	Fused Deposition Modelling
FEM	Finite Element Method
FJM	Fusion Jet Modelling
HEMS	Hybrid Electromagnetic Suspension
IPU	Inverter Power Unit
LIM	Linear Induction Motor
LPU	Levitation Power Unit
LSRM	Linear Switched Reluctance Motor
MOSFET	Metal-Oxide Semiconductor Field-Effect Transistor
NBR	Nitrile Butadiene Rubber
PCB	Printed Circuit Board
PCM	Phase Change Material
SLA	Stereolithography
SLS	Selective Laser Sintering
VCU	Vehicle Control Unit

Chapter 1

Introduction

The Hyperloop concept aims at providing fast, comfortable, and low-emission transportation. It makes use of a vacuum tube and electromagnetic suspension and propulsion as shown in Figure 1.1. By reducing drag, vibrations, and rolling resistance, it allows for speeds above 800 kph. Shortening one-day rail trips to an hour, it can potentially replace short- and mid-distance air travel.

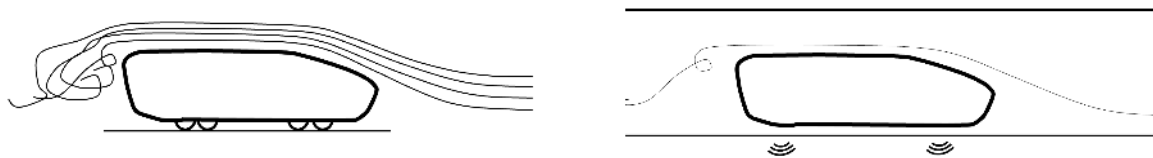


Figure 1.1: Left: Traditional Rail. Right: Hyperloop. Using a depressurized tube and electromagnetic levitation, the Hyperloop can reach high speeds at reasonable energy consumption.

There are, however, many technical issues to still be resolved. These include reliable and efficient levitation, linear electromagnetic motors, thermal management and cost-effective infrastructure. Various private organisations, research institutes and state-sponsored associations are working on these problems. Additionally, student teams compete internationally by building a working down-scaled vehicle over the span of one academic year. Prototypes and the developed technologies are displayed each summer at the European Hyperloop Week.

1.1 Swissloop

Swissloop [1] is a student-led team consisting of engineering students from ETH Zurich and other universities. Vehicle design, assembly and testing is done by students entirely. As an association, Swissloop operates to a large extent on sponsorships from private industry, ETH Zurich and EMPA (Eidg. Materialprüfungsanstalt). Currently, about 30 active members, half of which full-time, are involved at Swissloop.

Since 2016, every year has seen a new prototype vehicle with considerable technological progress. The prototypes built are scaled down significantly, in order to save cost and ease logistics. A scale comparison to a full-scale vehicle is shown in Figure 1.2. Figure 1.3 shows a recent prototype vehicle in action.

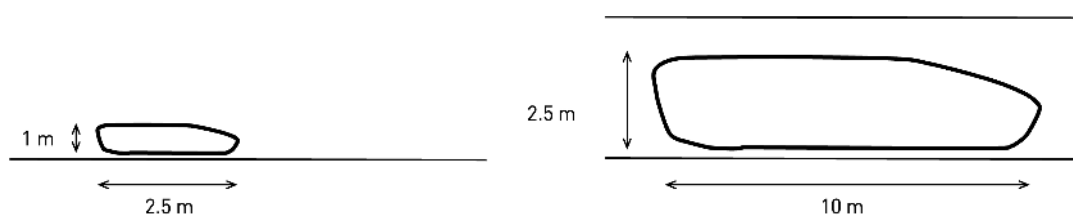


Figure 1.2: Left: Swissloop Prototype Scale. Right: Full-Scale Hyperloop Vehicle. The reduced size for prototype vehicles allows for affordable, yearly design iterations.



Figure 1.3: Swissloop Prototype "Simona de Silvestro" [2].

For the 2022/2023 season, the prototype will feature a linear switched reluctance motor tuned for efficiency and top speed. Additionally, active vertical and lateral levitation systems are in development. More minor technological improvements are pursued on almost every system of the vehicle, too. One challenge is the cooling of electronics in a near-vacuum environment. It is described in this bachelor's thesis in more detail.

1.2 Scope

To start, a brief overview of possible cooling concepts (convection, radiation, latent heat storage) is provided. Then, phase change cooling is examined in detail in this thesis. First, the potential of phase change cooling on a full-scale system is analyzed. Second, the simulation, design and manufacturing (along with some limited testing) of a small latent heat storage prototype is discussed in detail in Chapter 3. The built small-scale system can be used by Swissloop in prototype vehicles over the coming years. As a third step, the practical learnings can be used to formulate some recommendations for a large-scale phase-change cooling system (see Chapter 5).

Chapter 2

Concept

To begin, the conditions a full-scale vehicle would encounter are explained in detail.

2.1 Problem Setting

The envisioned vehicle will travel in a low-pressure environment of between 5 and 100 mbar [3]. Therefore, natural convection as a means of cooling is severely limited. With various on-board heat sources ranging from motors and power electronics to air conditioning, thermal management presents a key issue for the Hyperloop concept. In the following sections, the feasibility of various means of thermal management are discussed. First, a focus is put onto dissipation using convection and radiation. As a second step, various means of heat storage, like sensible and latent heat sinks, are discussed.

For the sake of this thesis, a full-scale vehicle with the following dimensions is used as a benchmark for various cooling solutions:

- Vehicle length: 10 m
- Vehicle height and width: 2.5 m
- Trip duration: 60 min (equivalent to Zurich - Rome or Zurich - Hamburg)
- Time between trips: < 10 min

On-board heat generation between 4 kW and 8 kW is assumed. Of this, 1 kW of heat generation is allocated for air conditioning for 10 passengers. 3 kW of heat is allocated for the levitation, and 4 kW for propulsion losses. Depending on the route, the propulsion is not always active.

For the small-scale prototype vehicle built by Swissloop in 2022/2023, the heat loss is 440 W for 15 minutes. Detailed boundary conditions (dimensions, power figures) for the small-scale prototype are discussed in detail in Section 3.1.

Goals

The goals of a cooling system with full-scale feasibility are three-fold. First, the electronics and passengers need to be kept sufficiently cool during high-speed travel runs. Second, this cooling must be repeatable. The vehicle should be able to do multiple runs a day without overheating, i.e. a sustainable form of heat management is necessary. Third, the time between runs should not be dictated by thermals. Here, it is prescribed that the time in between trips should not exceed 10 minutes.

2.2 Dissipation: Convection and Radiation

To achieve sustainable dissipative thermal management, a quasi-stationary state of heat transfer from the vehicle to its surroundings needs to be achieved. The three principal methods of stationary heat transfer are conduction, convection and radiation. Since the vehicles are envisioned to fully levitate, they will not touch any of their surroundings. As such, heat conduction is excluded from the outset. Below, convection and radiation are examined in more detail.

2.2.1 Convection

Dissipating heat through convection could be achieved as shown in Figure 2.1. Low-pressure air is channeled through a large array of fins, which in turn cools on-board electronics.



Figure 2.1: Low-pressure Convective Cooling. Using a fin array significantly increases the area available for heat transfer.

Theory: Gas Regime

To apply adequate correlations describing convective heat transfer, the flow regime the sparse air in the tube falls under must first be determined. The Knudsen number indicates the flow regime low-pressure gases abide by [4], as shown in Table 2.1.

Table 2.1: Flow Regimes of Gases. The Knudsen number expresses the importance of individual molecules on gas behaviour. The correlations describing convection vary drastically over these regimes.

	Kn	< 0.001	Continuum Regime
0.001	$< Kn$	< 0.01	Slip Flow Regime
0.1	$< Kn$	< 10	Transition Regime
10	$< Kn$		Molecular Regime

In the continuum regime, inter-molecular effects are relevant, and a fluid can be treated as a continuum. In this regime, fluids behave as is typical for gases around atmospheric conditions. Typical models for approximating convective heat transfer can be used [5]. On the other end of the spectrum, for example if pressure is low or the gas is confined to a small physical system, lies the molecular regime. Here, the behaviour of a gas is dictated by single molecules. For example, heat conduction between a vehicle and tube wall for $Kn > 10$ would not occur through a series of particle collisions in a continuous gas. Rather, individual particles would transfer heat by travelling between two walls directly, avoiding collision with other particles. In this regime, convective heat transfer models valid at atmospheric conditions fail.

The Knudsen number is defined as [5]:

$$Kn = l/L \quad (2.1)$$

where l is a molecules free mean path, and L is a system's characteristic length. The free mean path l for a gas is given by [6]:

$$l = \frac{k_B T}{\sqrt{2} \pi d^2 p} \quad (2.2)$$

where k_B is the Boltzmann constant, T is the temperature and p the pressure of the gas, and d the molecular diameter. To find the range of Kn for the specific problem at hand, T is set to 293.15 K. Furthermore, it is assumed that p is between 10 and 100 mbar, and that d for air is 4 Å. In the envisioned tube environment, the free mean path of air as formulated in Equation 2.2 would thus be between 647.8 nm and 6478 nm. To get an approximation of Kn , the characteristic system length L has to be set. Here $L = 0.5m$ is chosen, since this will be the length of the flow over the fin array.

Using Equation 2.1, Kn is calculated to be between $1.3 \cdot 10^{-5}$ and $1.3 \cdot 10^{-6}$. Therefore, the air in the shown system lies firmly within the continuum regime. All further computations for the convective heat transfer can thus be made using similar methods as used at atmospheric conditions.

Table 2.2: Material Values and Key Figures Used for Estimation of Convective Heat Transfer.

	10 mbar	100 mbar
μ	$1.81 \cdot 10^{-5} \text{ Pa s}$	$1.81 \cdot 10^{-5} \text{ Pa s}$
ρ	0.0119 kg/m^3	0.119 kg/m^3
λ [8]	0.008 W/mK	0.022 W/mK
c_p	1.003 J/kgK	1.003 J/kgK
Re_L	$0.73 \cdot 10^5$	$7.3 \cdot 10^5$
Pr	2.26	0.822
$\overline{Nu}$	39.3	6.00
$\overline{\alpha}$	$0.0959 \text{ W/m}^2\text{K}$	$0.264 \text{ W/m}^2\text{K}$
$\dot{Q}$	2.1 kW	5.8 kW

Theory: Convection on Fins

While convective skin cooling via the outer fairing could be possible, an array of cooling fins fed through an air inlet is used as a basis for computations here. A central fin array can be placed physically close to heat sources, reducing issues with integration and heat transfer. It is assumed that the fin array (similar to an automotive radiator) is 2 m in width, 1 m in length, and 0.5 m in height at most to fit into the vehicle. The fin array consists of 100 stacked aluminium sheets of 1 mm thickness each, making for a surface area A of 400 m^2 . Perfect heat conduction is assumed in the metal. The convective flow is assessed as a laminar flow over a planar surface according to [7].

The density shown in Table 2.2 is calculated from the ideal gas law at 293.15 K. The dynamic viscosity μ of air is assumed to be exclusively dependent on temperature. Shown below are the material properties used in the following calculations.

First, the occurrence of turbulence is examined. The Reynolds number at the flow exit ($L = 0.5 \text{ m}$) is given by:

$$Re_L = \rho \cdot \frac{u_\infty L}{\mu} \quad (2.3)$$

where u_∞ is the incoming flow velocity of 222 m/s (800 kph). Since $Re_L > 10^5$ for the 100 mbar case, the flow is expected to be turbulent. For the case of 10 mbar tube pressure, the flow is laminar. Next, the Prantl number is dependent on heat capacity c_p , thermal conductivity λ , and dynamic viscosity μ .

$$Pr = \frac{c_p \mu}{\lambda} \quad (2.4)$$

The results for both Prantl and Reynolds number are both noted in Table 2.2.

Theory: Turbulent Convection

For the turbulent case (100 mbar of tube pressure), the local Nusselt number is given by:

$$Nu_x = 0.0296 \cdot Re_x^{4/5} \cdot Pr^{1/3} \quad (2.5)$$

with Re_x being the local Reynolds number. Derived from this, the average Nusselt number over the fin length can be found using [7]:

$$\overline{Nu} = 0.0296 \cdot Pr^{1/3} \cdot \left(\frac{u_\infty \rho}{\mu}\right)^{4/5} \cdot \int_0^L x^{4/5} dx \quad (2.6)$$

From $\overline{Nu}$, the length-averaged heat transfer coefficient $\overline{\alpha}$ can be found [7]:

$$\overline{\alpha} = \overline{Nu} \cdot \frac{\lambda}{L} \quad (2.7)$$

and, finally, from $\overline{\alpha}$ the possible heat flow from convection:

$$\dot{Q} = A \cdot \overline{\alpha} \cdot \Delta T \quad (2.8)$$

with ΔT being the temperature between metal fins and the gas in the tube. Here, the gas temperature is set to 20 °C. Furthermore, it is assumed that for efficient operation the temperatures of on-board electronics are not allowed to exceed 80 °C during steady state operation. Due to some small thermal resistances, it is assumed that the fins have a temperature of 75 °C. ΔT is thus 55 °C.

The expected heat flow from convection at 100 mbar ($\overline{Nu} = 6$) using equations 2.3 to 2.8 is 5.8 kW. Relevant intermediary values are noted in Table 2.2. This means that, for the higher end of feasible tube pressures, convection can be a sufficient means of heat dissipation. At 100 mbar, the cooling power is 6.8 kW. There is, however, significant drag associated with the use of a large fin array. Additionally, it would be heavy at a mass of 540 kg for the aluminium sheets exclusively, not yet factoring in additional hardware. Lastly, for lower pressures the heat transfer significantly decreases, as shown in the following section.

Theory: Laminar Convection

Since $Re_L < 10^5$, a different correlation for Nu_x is used here [7] and rewritten to a similar form as Equation 2.6:

$$\overline{Nu} = 0.332 \cdot Pr^{1/3} \cdot \sqrt{\frac{u_\infty \rho}{\mu}} \cdot \int_0^L x^{1/2} dx \quad (2.9)$$

In the laminar case at 10 mbar, the power which can be dissipated using a fin array is only 2.1 kW. Again, intermediary steps are note in Table 2.2. This decrease from 100 mbar is caused by following effects:

- Due to the flow being laminar, heat transfer to the fluid decreases.
- The incoming air has low density and thus low heat capacity.
- Thermal conductivity is significantly lower.

All of these effects trace back to the low pressure.

Conclusion

At higher pressures, where propulsion concepts based on air turbines is being conducted [9], convective heat transfer can be sufficient. While 5.8 kW of dissipation is in the desired range, extensive research has to be put into optimizing the ratio of heat transfer to drag of a large fin array.

At 10 mbar, the dissipative performance relative to weight, drag and space usage is underwhelming. With this, convection is ruled out as a suitable means of heat dissipation at lower pressures approaching 10 mbar. For all further investigation, focus is put on this low pressure (10 mbar) scenario.

2.2.2 Radiation

Radiative heat transfer depends on the geometries and materials used in vehicle and tube. Still, a simple model is used to assess radiation. Ideal values are assumed wherever possible to determine an upper limit of cooling through radiation.

Theory: Cylinder Approximation

To this end, the vehicle geometry is approximated as a cylinder as shown in Figure 2.2. The radiative heat transfer between an inner and outer hollow cylinder is given by [10]:

$$\dot{Q} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \left(\frac{A_1}{A_2}\right) \cdot \left(\frac{1}{\varepsilon_2} - 1\right)} \quad (2.10)$$

where subscript ₁ represents the inner cylinder (vehicle) and subscript ₂ the outer cylinder (tube). The area A relates to model depth and circumference, meaning the flat ends of cylinders are not involved here.



Figure 2.2: Left: Radiative Dissipation on Outer Fairing. Right: Substitute Model. The substitute model represents the pod as a cylinder.

The Stefan-Boltzmann constant is given as:

$$\sigma = 5.67 \cdot 10^{-8} \frac{W}{m^2 K^4} \quad (2.11)$$

The pod emissivity ϵ_1 and tube emissivity ϵ_2 are assumed to be equal to 1. The area of the vehicles outer fairing A_f replaces A_1 , rendering the following equation:

$$\dot{Q} = A_f \sigma (T_f^4 - T_t^4) \quad (2.12)$$

with T_f being the temperature of the vehicle fairing and T_t being the temperature of the tube. Conservatively, T_t is set to 40 °C to represent outside air on a European summer day.

Furthermore, as before it is assumed that the maximum allowed electronics temperature is 80 °C. Due to thermal resistances between electronics and fairing, these electronics are likely to be significantly warmer than the outer fairing. Thus, it is assumed that the outer fairing can be held at a maximum of 65 °C. With a vehicle length and diameter of 10 m and 2.5 m respectively, and assuming that roughly 50 % of the surface area can be used for radiation, a radiative heat transfer of 7.7 kW can be reached.

It must be noted, however, that elevated outer fairing temperatures raise some issues. First, the vehicle must be well-insulated from the hot outer fairing. Otherwise, inadvertent back-flow of heat into sensitive components and passenger cabin can occur. Second, sufficient in-plane heat conductivity is necessary in the fairing. Otherwise, the fairing's in-plane thermal resistance can easily act as a bottleneck. In this case, a given needed heat flow is associated with large temperature gradients over the fairing, leading to excess temperatures at the heat sources. Enabling in-plane heat flow poses a significant engineering challenge, especially with weight and aerodynamics as competing goals.

On the other hand, $\dot{Q} \propto T_f^4 - T_t^4$ is a quartic relationship of heat flow and temperature. Therefore, slightly raising fairing temperatures by decreasing thermal resistance vastly enhances radiative heat dissipation. For example, bringing T_f from 65 °C to 70 °C increases $\dot{Q}$ by 23 %.

This leads to another possible method of radiative heat transfer. Using a heat pump, a probe can be heated to very high temperatures. By thermally insulating it from the vehicle, it can dissipate heat with very small surface area. The concept is shown in Figure 2.3.

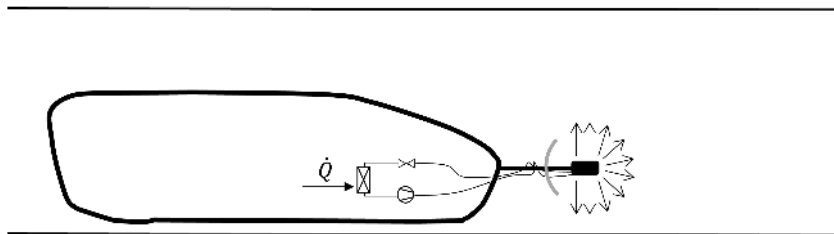


Figure 2.3: Radiative Dissipation with Heat Pump and Probe. Using the heat pump, high temperatures in the probe can be achieved, which drastically increases radiative heat transfer.

Conclusion

In this example, with 7.7 kW being firmly in the 4 kW - 8 kW range, enough cooling can be provided. With radiative heat transfer being independent from air as the heat transfer medium, the cooling performance is independent of tube pressure. Thus, to conclude, it can be stated that radiation has clear potential

as a cornerstone of thermal management in Hyperloop vehicles. However, only a brief assessment has been conducted with two possible approaches stated. Further research is thus needed.

Furthermore, it is impossible to test radiation in everyday usage on small-scale prototypes during summer outside of a tube. The high amount of radiative heat input from the sun would render a radiative cooling system unusable. Thus, it is much harder to discover operational flaws of a prototype system, so that the idea was not pursued any further. Following below, focus is shifted from dissipation of heat to another promising form of thermal management: storage. Through discharging rapidly between runs, storage can become operationally sustainable. First, sensible heat storage is looked at.

2.3 Sensible Heat Storage

Sensible heat storage occurs when a material's temperature raises as it is supplied with heat. Generally it holds:

$$\dot{Q} = c_p \cdot m \cdot \Delta T \quad (2.13)$$

In the past, sensible heat storage has been used by Swissloop on prototype vehicles. Aluminium, due to its low cost, high c_p and λ , makes for a suitable heat sink for short, high-power applications.

2.3.1 Aluminium Heat Sink and Other Metals

However, for usage in longer timeframes, sensible heat storage is not suitable due to its low thermal capacity. The maximum allowed ΔT is set to 40 °C. Using Equation 2.13 with aluminium (888 J/kgK), and integrating over time, one obtains:

$$m = \frac{t \cdot \dot{Q}}{\Delta T c_p} \quad (2.14)$$

At 8 kW over a time t of one hour, the mass for an aluminium heat sink is equal to 810 kg. At 440 W over 15 minutes (Swissloop Prototype), this figure is reduced to 11.15 kg. For reference, the latent heat storage module shown later needs less than 1.4 kg of PCM to reach a similar capacity. There are no metals with higher heat capacity than aluminium known to the author. With this, metals are discarded as a primary heat storage device for the given application.

Gases were previously discarded as sensible heat storage materials due to volumetric constraints in the envisioned usage scenario. Among solids, there are few materials (ceramics, polymers, ...) with higher specific heat capacity than aluminum. When it comes to fluids, however, some perform very well.

2.3.2 Water

Water, and some fluids used with it in automotive and rail applications, such as alcohols (e.g. glycol), have a very high specific heat capacity. At 440 W for 15 minutes with $\Delta T = 40$ °C, 2.36 kg of water would provide the needed thermal capacity. (172 kg at 8 kW for 60 min.) Water is easy to handle, can be run directly through heat exchangers, and is cheap. Considering this, it seems like a suited heat storage medium for the given application. However, water is still 40 % heavier than PCM with equivalent capacity at $\Delta T = 40$ °C. If the tolerable ΔT were smaller, PCMs would outperform water by an even larger margin.

2.3.3 Combined Structural and Thermal Usage

Other than using water, another potential way of improving the mass-specific capacity of sensible heat storage is simultaneous structural and thermal usage of components. With Aluminum (888 J/kgK) and carbon fiber reinforced plastics (CFRP, 1040 J/kgK) making up a large fraction of structural materials in mobility applications, there is a high likelihood that the needed thermal mass on a high-speed vehicle would already be present in structural components. Figure 2.4 shows a component as typically used in Swissloop vehicles. This component is optimized for structural loads. With few adaptations, however, it could serve as a heatsink.

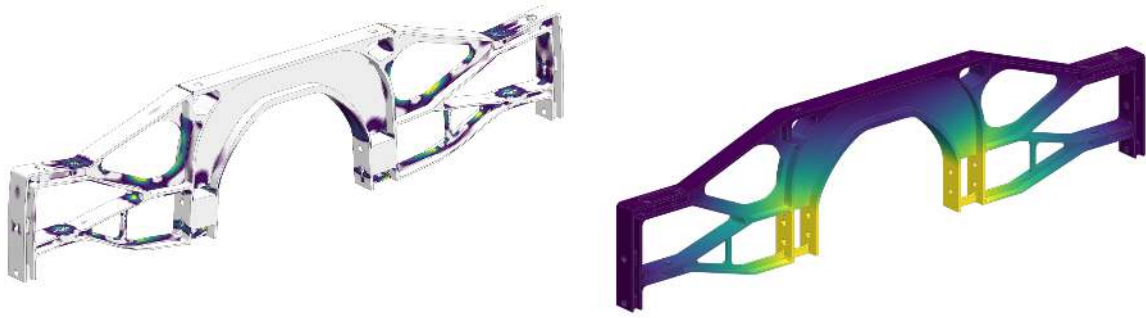


Figure 2.4: Structural Vehicle Component Used as Heatsink. Left: Stresses, Right: Temperatures. Simultaneous structural and thermal usage allows for weight savings.

Both CFRP and aluminium do not experience a loss in structural performance in the temperature range between 0 °C and 80 °C. Therefore, if thermal expansion can be accounted for, a ΔT of 40 °C can easily be tolerated by structural components used as heatsinks. Some hardware might be needed to ensure heat transfer from electronics into structural parts. However, the capacity weighs virtually nothing since no material needs to be added for capacity. Like water, this seems an ideal solution at first glance.

Heat Transfer and Entropy

However, there are some issues with such a solution:

1. Structural considerations can lead to geometries not optimal for heat transfer (e.g. thin walls). Optimizing for both criteria comes at the cost of some additional mass.
2. Large structural components spanning a whole vehicle are generally less suited for good heat transfer. Their large dimensions lengthen the effective path of heat flow, increasing temperature gradients. There might be a need for additional engineering measures to ensure that all electronics are kept cool.
3. With the warming of structural components, heat leakage into other parts of the vehicle tends to increase. Thus, for combined thermal-structural usage, a relevant amount of heat will be released into unwanted areas (e.g. passenger cabin). A more centralized metal heatsink or water tank would perform better in this regard, but might still require significant amounts of insulation.
4. Similarly, an increase in entropy is associated with warming structural components and consequent heat leakage. This would make it virtually impossible to retrieve this heat between consecutive runs. Thus, using vehicle structures as heat sinks can not be seen as a sustainable cooling concept for large-scale applications.

2.3.4 Conclusion: Use as Complementary Storage

It has been shown that sensible heat storage such as water or metal heat sinks can be lacking in terms of mass-specific thermal capacity despite large ΔT . Still, these solutions come with some benefits:

1. Water, and other fluids, can be pumped. Thus, retrieving warmed water from a vehicle tank is rather simple. As such, water is very suitable for rapid retrieval between consecutive vehicle runs.
2. Both water and aluminium make for great heat transfer media. Water, through its ability to be pumped, can be heated very homogeneously at a heat source. The result is good heat transfer and in consequence low electronics temperatures. Similarly, aluminium has a high thermal conductivity. Thus, both are well-suited for applications with relatively large bursts in heat.

The second point makes these materials a great choice as intermittent heat sinks. They can be used to dampen power peaks during usage, and as heat transfer media. Both water and aluminium are extensively used in the small-scale prototype described later. To achieve greater mass-specific capacity, the solution proposed below is using phase change materials.

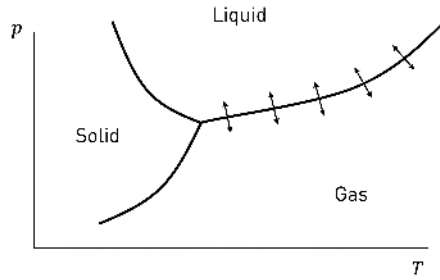


Figure 2.5: Typical p-T Phase Change Diagram. The temperature of a material's liquid-gaseous phase change is pressure-dependent.

2.4 Phase Change (Latent Heat Storage)

Three modes of phase change were considered: Liquid-gaseous phase change, solid-solid phase change, and solid-liquid phase change.

2.4.1 Liquid-Gaseous Phase Change

Of these three, liquid-gaseous phase change (evaporation) can by far achieve the highest mass-specific enthalpy of phase change. For evaporation, the temperature of phase change is highly dependent on pressure, as shown in Figure 2.5. This allows for a tunable temperature of phase change by adjusting the pressure a liquid/vapor mixture is under.

Table 2.3 shows various options that were considered for use as phase change media. At a pressure of 1 bar, which can be kept inside a vehicle without specialized hardware, various alcohols have a suitable temperature of phase change. These include methyl acetate and methyl iodide, chloroform, acetone, carbon disulfite and many more. As an alternative approach, gases can be kept at high pressures (e.g. ammonia) or at low pressures (e.g. water) to achieve the desired phase change temperature. However, for all of these options, the volumetric expansion inherent to liquid-gaseous phase change poses severe issues. First, keeping constant pressures at changing volumes in an isolated system is challenging and requires heavy and space-consuming parts, like pumps or bladders. Second, as seen in Table 2.3, liquid-solid phase change is volumetrically inefficient.

To illustrate this point, the volumes of a solid-liquid and liquid-gaseous PCM on the Swissloop prototype are computed. At roughly 300 kJ of storage capacity, 1.5 kg of solid-liquid PCM with 200 kJ/kg is required. When evaporating water at 80 mbar with an enthalpy of phase change equal to 2403 kJ/kg, the density of the fully evaporated water is 18.1 m³/kg [11]. The resulting volume is roughly 2.3 m³. For a small prototype vehicle of 2.5 m x 1 m x 1 m, the phase change thermal management unit would thus consume almost all of the available space in the vehicle. In a full-scale application, the large volume fraction of a passenger cabin makes space concerns more relevant. Thus, the volume occupied by a liquid-vapor PCM is seen as even less feasible in these scenarios. While some pressurized gases, like ammonia, perform slightly better, their volume-specific capacity is still lacking. All together, a liquid-gaseous phase change in a closed system is ruled out.

Table 2.3: Volume Usage and Mass-Specific Capacity of Liquid-Gaseous PCMs. The large volumes occupied by the gaseous PCMs (v_{vap}) make them volumetrically ineffective compared to a solid-liquid PCM like paraffin.

	h_{pc}	Pressure	T_{vap}	v_{vap}
Carbon Disulfite [12]	355 kJ/kg	1 bar	46 °C	$\sim 0.35 \text{ m}^3/\text{kg}$
Methyl Acetate [13]	405 kJ/kg	1 bar	57 °C	$\sim 0.35 \text{ m}^3/\text{kg}$
Ammonia [11]	1094 kJ/kg	16 bar	41 °C	$0.09 \text{ m}^3/\text{kg}$
Water Vapor [11]	2393 kJ/kg	0.1 bar	47 °C	$14.67 \text{ m}^3/\text{kg}$
Paraffin RT 43 HC	240 kJ/kg	1 bar	43 °C	$0.0013 \text{ m}^3/\text{kg}$

As an alternative to a closed system, the evaporated gases can be vented into the vacuum tube during operation. From a perspective isolated to the vehicle, this represents an ideal solution. However, this approach comes with extensive measures which have to be taken to retrieve these vapors from the tube environment over large volumes. For sustainable operation, the evaporated liquids need to be retrieved using specialized pumps. Hence, the option of liquid-gaseous phase change is discarded.

2.4.2 Solid-Solid Phase Change

Solid-solid phase change materials are not as commercialized as liquid-solid PCMs. There are, however, some advantages to these "solid state" PCMs [14]:

- Small volume change during phase change
- No need for encapsulation

For the few commercially available solid-solid PCMs, the mass-specific capacity was found to be significantly lower than for salt- or paraffin-based PCMs. Typical enthalpies of phase change for solid-solid PCMs are roughly 100 kJ/kg [15], compared to double that for solid-liquid PCMs. Additionally, density and thermal conductivity are both in similar ranges as paraffin and salt hydrate PCMs. While ongoing research does not exclude future usage of solid-solid PCMs, it was concluded that the drawbacks of commercially available solutions outweighed the advantages. Therefore, these were discarded as well.

2.4.3 Solid-Liquid Phase Change

With solid-solid and liquid-gaseous phase changes discussed, a third available option is employing solid-liquid phase change materials.

Water/Ice

One option for a solid-liquid PCM is water, i.e. the melting of ice. This phase change comes with a very high enthalpy of phase change at 333 kJ/kg. Since this phase change occurs at 0 °C, operating a latent heat storage unit would pose issues of unwanted melting. In particular, during usage on the Swissloop prototype at atmospheric pressure during summer, melting will occur at significant rates even when the vehicle is not in use. To combat this, significant amounts of insulation are needed to prolong the duration of passive melting. Similarly, in a full-scale vehicle in vacuum some leakage inevitably occurs, for example from a heated passenger cabin.

For a prototype vehicle, this means that using a water-based PCM cooling system is associated with impractical handling concerns. For example, the vehicle can not be left sitting for too long between test runs, which is a major inconvenience during testing. Water as a phase change material was thus discarded for the system being built here.

In a full-scale prototype, water could be investigated further, as it does come with some advantages. Melted water is easy to pump, and also has a high sensible heat capacity c_p . For such an application, the amounts of insulation needed would have to be examined in more detail to determine full-scale feasibility.

Hydrated Salts

Salt hydrates pose one alternative to water. Typically, a crystalline solid of the form $AB \cdot xH_2O$ is formed from inorganic salts (AB) and water [16]. Here, the "melting" occurs through dehydration of the salt. Salt hydrates usually melt to a salt hydrate with fewer moles of water, or to its anhydrous form. During this transition, liquid water released from the hydrated salt dissolves the formed nonhydrated salt molecules [16]. For commercially available salt hydrate PCMs, the maximal operating temperatures are limited to roughly 20-30 °C above melting temperature [17, 18].

Salt hydrates are affordable and widely available. They are, for example, commonly used in hand warmers. However, salt hydrates can lack cyclical stability. Depending on manufacturing quality and composition, the performance of hydrated salt PCMs can decrease drastically over time.

A large disadvantage of hydrated salt PCMs is that, depending on composition, they can be corrosive to metals. A circumstance which, if metal heatspreaders are planned to be used, requires additional engineering to prevent long-term corrosion. Hydrated salts come with improved density over water and

other solid-liquid PCMs, and are thus preferable in volume-critical applications. However, the mass-specific enthalpy of phase change is lower than with other PCMs, which can be a drawback. Table 2.4 compares the key performance characteristic of a typical hydrated salt PCM with a paraffin alternative.

Paraffin

Paraffin (from latin "parum affinis", little reactivity) typically consists of multiple alkane types with the form C_nH_{2n+2} . For paraffins, the hydrocarbon chain length expressed by n influences the melting temperature and enthalpy of phase change. Generally, more pure compositions limit the temperature of phase change to narrow temperature range.

Paraffin is a by-product of the petro-chemical industry and is commonly used in candles, vasoline, and similar everyday products. Due to well-established, large-scale manufacturing processes, paraffin in higher grades is affordable and can be purchased from a variety of vendors. As a PCM, paraffin is available at a wide range of phase change temperatures and possesses high specific enthalpies of phase change. Additionally, it shows high cyclical stability, meaning melting and subsequent solidification can be repeated over a long lifespan [19]. Generally, more consistent quality is expected from paraffin PCMs across manufacturers and their product ranges. This is not the case with salt hydrates.

Both paraffin and hydrated salt PCMs suffer from low thermal conductivities [20]. In many applications, the resulting temperature gradients in the PCM can cause insufficient cooling at the heat source. Thus, measures have to be taken to ensure adequate heat transfer. Very commonly, the PCM is placed in a metal enclosure with fins.

Table 2.4: Typical Hydrated Salt and Paraffin PCMs.

	RT 35 HC [19]	SP 31 [17]
T_m	35 °C	31 °C
h_{pc}	210 kJ/kg	180 kJ/kg
λ, k	0.2 W/mK	0.5 W/mK
ρ_s	0.88 kg/L	1.35 kg/L
	Inert	Corrosive

While paraffin achieves slightly higher mass-specific capacity (Table 2.4), hydrated salts are both denser and have significantly higher thermal conductivity. In effect, higher density also contributes to elevated heat capacity, since the path length heat has to travel is reduced. Thus, it is likely that hydrated salt PCMs require less heatspreader material at given heat flows. Thus, despite lower mass-specific capacity, the overall system is likely to be lighter than with paraffin. Figure 2.6 shows this idea graphically. The paraffin heatspreader is significantly larger and thus heavier, offsetting the weight advantage of paraffin.

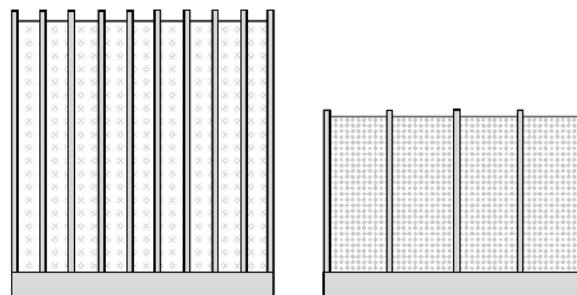


Figure 2.6: Size Comparison of Heatspreader for Paraffin and Salt Hydrate PCM. Left: Paraffin-based PCM with heatspreader. Right: Salt hydrate PCM with heatspreader. The density and thermal conductivity of salt hydrates allow for a smaller heatspreader with fewer fins.

Despite the volume and weight advantages of salt hydrates, the decision was ultimately made to employ a paraffin PCM here. Two major factors played into this choice. First, to avoid corrosion of the

PCM container, the chemically inert paraffin is advantageous. Second, the reliability concerns for salt hydrates meant that only one manufacturer (Rubitherm Technologies, Berlin) was taken into consideration for this PCM type. However, their limited product range meant that no salt hydrate with a suitable temperature of phase change could be found.

2.4.4 Improvement of Thermal Conductivity

With paraffin as the PCM of choice, a significant issue is limiting the mass of metal needed for an adequate heatspreader at given heat transfer goals. To this end, additives such as alumina powder, expanded graphite or short-strand carbon fiber are commonly used in PCMs.

Existing Research

There is extensive research on this topic. A few relevant insights are highlighted here, with focus on additives used for hydrated salt and paraffin PCMs.

Fukai et al. [21] experimented with carbon fibers of lengths 5 mm to 200 mm in paraffin and hydrated salts. They found that careful orienting of the fibers can enhance thermal conductivity, and characterized the relationship between carbon fiber volume fraction and composite thermal conductivity. To give an example, they found that 2 % carbon fiber can increase the thermal conductivity of paraffin six-fold to 1.2 W/mK.

Karaipekli et al. [22] compared carbon fiber and expanded graphite additives in stearic acid. They found that expanded graphite (and to a lesser degree carbon fiber) does not separate from the PCM during repeated melting. Despite expectations that these additives would settle due to their higher density, they were cyclically stable.

Similarly, Frusteri et al. [20] demonstrated a stable suspension for short-strand carbon fibers in a hydrated salt PCM. Analog to paraffin and fatty acid based PCMs, correlations between volume fraction of additives and thermal conductivity could be found. Frusteri et al. also note that homogeneous distribution of additives in the PCM affects results.

Using extensive computational fluid dynamics (CFD) studies, Shatikian et al. [23] studied the effect of metal fin dimensions on the melting of paraffin PCM. These studies account for natural convection caused by buoyancy of melted paraffin, which has lower density than in solid state. The modelling of melting and convection of PCMs is discussed in detail below.

Overall, extensive research on the topic has been conducted. For both RT 35 HC and RT 44 HC, the manufacturer claims compatibility with graphite additives. These additives, however, cause increased viscosity and potentially impede the fill-in procedure of the PCM. Employing additives in the prototype cooling module was briefly considered, but later discarded due to time constraints. For future versions of the prototype PCM cooling module or in full-scale applications, employing additives should be considered.

Chapter 3

Design for Prototype

This chapter details the design of a complete PCM-based cooling system. Built with the small-scale Swissloop prototype of 2022/2023 in mind, the system will see use until summer 2023 with use at the European Hyperloop Week. Depending on performance, the system might be re-used again in coming seasons. Or, if areas of improvements are found, a second version can be designed. Due to the time frame Swissloop operates under, production of the newest prototype vehicle has only just started at the time of writing. Additionally, some of the sensors, in particular those for pressure and pump speed, will only be read once on-board electronics are operational. Thus, testing under real-world conditions can only be conducted after work on this thesis has concluded. Definitive conclusions relating to reliability, warm weather resilience, and full-load cooling performance can only be drawn about five months after submission.

3.1 Boundary Conditions

The 2022/2023 Swissloop prototype vehicle has two major operational modes: The default mode and the levitation mode. The default mode consists of fast accelerations using the linear switched reluctance motor, followed by hard mechanical braking. Recuperative braking can be activated, too. These runs take 10 seconds at most. Simultaneously, the lateral and vertical levitation systems are activated. These consist of magnetic coils with active current control. By regulating attraction forces in the left/right and up/down directions, constant airgaps to the track can be held. In the default mode, the levitation is activated before accelerating. It is, however, not active for much longer than a minute at most. In the default mode, long gaps between runs are unavoidable. Between every run, vehicle resets and inspections are conducted, lasting at least a minute. In practice, 15 runs can be conducted per day at most during testing or demonstration.

In the pure levitation mode, the vehicle is electromagnetically suspended on a short piece of track for up to 15 minutes. This mode can also be performed in a vacuum chamber. In both modes, the majority of heat losses is concentrated to a few sources.

At the time of writing, an additional full speed mode was considered to be implemented. The heavy electromagnets for levitation and other non-essential hardware is removed, making for reduced vehicle mass. The lower mass allows for higher accelerations, and thus testing of the motor at higher speeds. The cooling system has not been built for this third mode.

Heat Source: Battery

The large Lithium-Polymer battery (800 V, 250 A) is only used in the default mode, since it is not rated for use in low pressure environments. It is assumed that in full-scale applications, the battery is a heat source which requires cooling. In the future, in particular for long operations in vacuum, this subject certainly needs to be tackled. However, it has been decided to not cool the battery on the prototype. Ordering a battery that would allow for tab cooling would exceed the production time frame, making it impossible to build a working battery system in a year. It has also been found that on the prototype, during the default mode, the battery mass can easily absorb the heat generated through internal resistance.

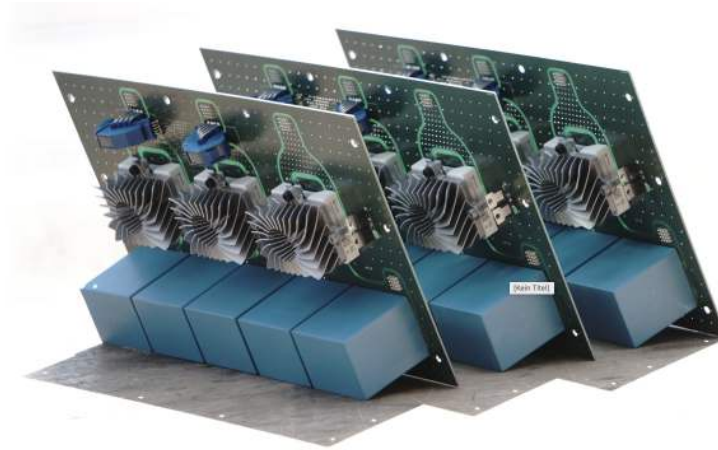


Figure 3.1: 2021/2022 Inverter Power Boards with Air Cooling [27].

Heat Source: Electromagnets

The second major heat sources are the electromagnets in the motor and levitation system. At the heart of the motor, there are cores of finely laminated electrical steel. These reduce magnetic losses from eddy currents. Around the cores, multiple layers of copper wire are wound. As in past years, the majority of losses in the motor are ohmic losses in the wires [24]. The motor is built for 150 A of peak pulsed current. During the default mode, these wires have been found to not exceed critical temperatures. Since the motor is active for less than ten seconds, adiabatic calculations for the wires were used. During the levitation mode, the motor is not in use. The levitation, too, is built around steel cores, which are CNC milled. These do not possess any lamination, since the magnetic flux is approximately static even when the vehicle is moving at high velocities. The copper windings of the levitation systems reach their highest temperature in the levitation mode after 15 minutes. On this timescale, the steel core acts as a heatsink for the windings. Thin epoxy coatings on the wires act as a thermal barrier. A brief transient simulation using FloEFD has led to the conclusion that the levitation system reaches temperatures of up to 90 °C after 15 minutes. These values are not problematic.

Like with the battery, it is evident that eventually these electromagnetic actuators must be actively cooled for continuous use at near-vacuum. Thus, implementing active cooling of the levitation system was initially considered as a proof of concept. Specifically, removing heat by using water as a heat transfer medium was looked at. This could be achieved by using hollow copper windings [25] or incorporating polymer tubing into the windings. As an alternative, the steel core could be fitted with internal water channels, or an external heatspreader could be pressed onto the windings. Because the levitation system is in its first iteration, its design is associated with many unknowns. Active cooling was thus discarded to focus on essential functionality.

Heat Source: Power Electronics

The power electronics link battery and electromagnets. For the motor, the power electronics are historically called "Inverter" at Swissloop. In the 2022/2023 vehicle, there are three inverter boards as shown in Figure 3.1. Per inverter board, three large MOSFETs [26] (half-bridge) are the centerpiece of the power electronics. The heat losses in these MOSFETs consist of switching losses and on-resistance. A high switching frequency is needed to accurately control current in the motor. However, the variability of the switching frequency means the heat loss can not exactly be predicted in advance. At a switching frequency of roughly 20 kHz and temperatures of 60 °C, these losses are expected to be around 150 W. This is assumed to be the heat loss going forward.

With the motor only being active in default mode, a total of nine inverter MOSFETs produce 150 W for 10 seconds simultaneously. The Levitation Power Unit (LPU), consists of four PCBs constructed similarly to the Inverter PCBs. Here, two smaller MOSFETs [28] (full-bridge) per PCB are expected to produce 50 W in heat losses each. An overview of the heat sources is provided in Figure 3.2.

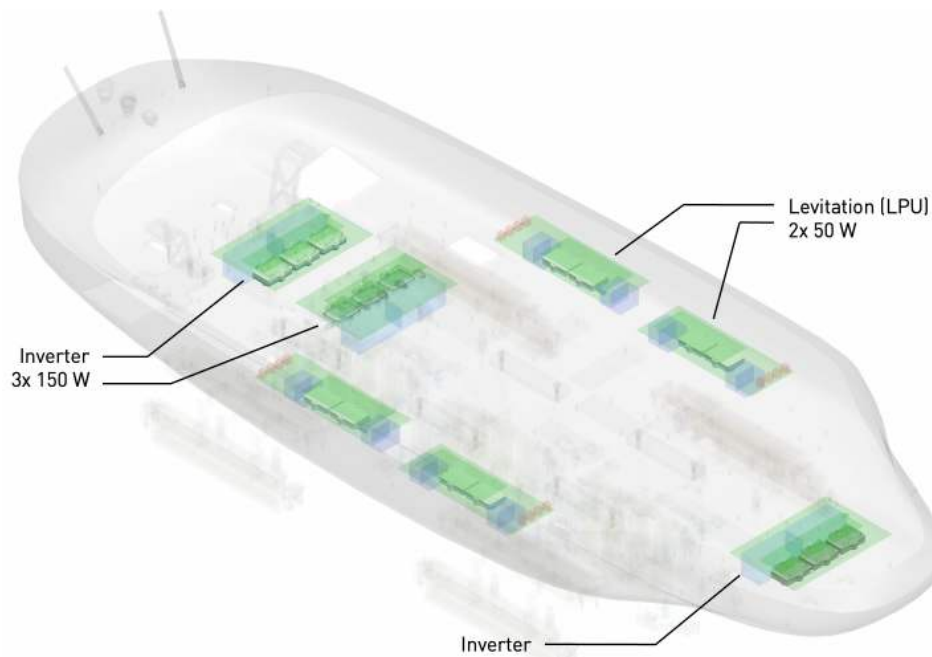


Figure 3.2: Heat Sources on 2022/2023 Swissloop Vehicle.

Active cooling in these power electronics is indispensable. As alluded to, past years have seen sensible heat storage (aluminium heat sinks) and air cooling as practical solutions. The LPU and Inverter MOSFETs have different footprints, but use identical mounting hardware. Thus, it is possible use a single heatsink design for both.

System Requirements

To summarize, the cooling system is built around the following power profiles:

- Handle multiple 10 s bursts of 9x 150 W and 8x 50 W (default mode).
- Dissipate 8x 50 W over 15 minutes (levitation mode).

Here, the levitation mode dictates the needed energy capacity of the PCM for the cooling system. At four boards with two MOSFETs each, a total number of eight MOSFETs produce 400 W over 15 minutes. Conservatively, this figure was later assumed to be 440 W. This exceeds the overall energy dissipated by 15 default runs occurring in a single day.

The following additional requirements were set early on for the cooling system:

- The complete cooling system should not exceed a weight of 7 kg, since pod acceleration and levitation are highly sensitive to vehicle mass.
- Temperatures of up to 35 °C should not impede the functionality of the cooling system.
- The power boards are mounted with the MOSFETs facing downwards. Thus, they are not easily accessible. The cooling system can not rely on easy accessibility of the MOSFETs.
- Structural integrity, in particular for use in levitation mode at near-vacuum needs to be guaranteed.
- Safety hazards for other pod components and human operators should be considered and minimized.

3.2 System Architecture

A major decision early on in the design process was the fundamental system architecture. Roughly, it was known how much PCM was needed to absorb the produced heat. However, two options were identified as to the distribution of that PCM: Decentralized or centralized. Figure 3.3 shows these two approaches. In the figure, a water loop is shown in dark blue. The patterned material represents a PCM, and grey represents metal.

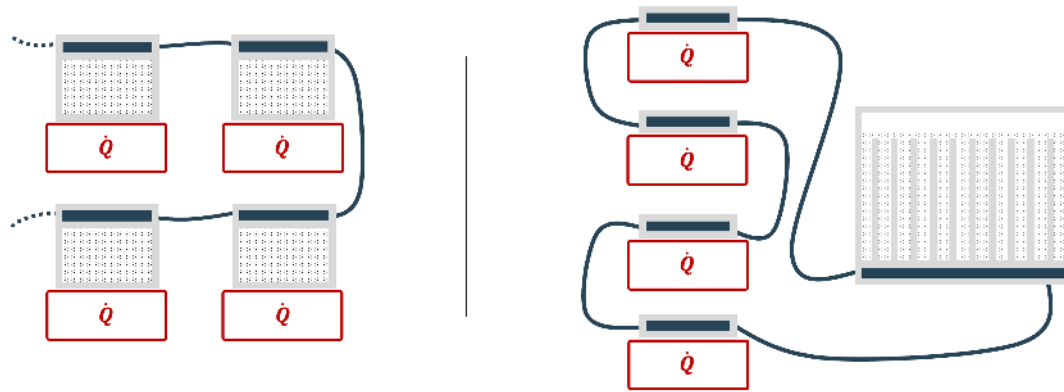


Figure 3.3: Left: Decentralized Architecture. Right: Centralized Architecture. During operation, the decentralized architecture employs the water loop for heat transfer.

3.2.1 Decentralized Architecture

The decentralized architecture approach centers around placing PCM as physically close to the heat sources as possible. The major advantage of this approach is good heat transfer, as the thermal path from source to PCM is short. Since these decentralized parts are not well accessible, tubing connects them. In-between runs, they can be flushed with coolant from an external pump. With the pump not on the vehicle, some weight is saved.

There are, however, multiple disadvantages. First, space restrictions on the PCBs might make it impossible to fit enough PCM into a decentralized container. All of these require tubing ports, with water chambers for flushing, a PCM chamber, and potentially heat spreaders. All of these add to considerable volume for each decentralized PCM container. The second, more restricting downside, relates to design flexibility. Using the decentralized approach, the designed solution can only ever be used for power electronics. If, for example, the electromagnets need to be cooled, a completely separate new design is needed.

3.2.2 Centralized Architecture

With regards to flexibility, a centralized architecture is hugely advantageous. Once the PCM container (here: module) is built, it can be used for various purposes. Generally, it is assumed that designing a fluid heat spreader at a heat source requires less specific knowledge than designing a PCM module. Thus, a centralized module can be used in future vehicles, with only adaptations to the fluid circuit. In the development cycle at Swissloop, this flexibility is highly valuable for coming vehicle designs. It allows for the centralized module to be used in future prototypes without a complete redesign, independent of MOSFET type and electromagnet dimensions. Thus, the centralized architecture was chosen here.

As an additional benefit, the centralized approach allows for more efficient usage of the available PCM material. During the default mode, for example, the Inverter MOSFETs produce a majority of the dissipated energy. Meanwhile, in the levitation mode, the LPU MOSFETs produce the majority of dissipated energy. In the decentralized approach, the needed PCM capacity on these MOSFETs is not fully used. The centralized approach allows for full usage of capacity independent of use case, reducing the needed amount of PCM.

A disadvantage of the centralized approach is inferior heat transfer. As a partial solution to this issue, the fluid loop and metal heat spreaders on the MOSFETs are used as intermediary buffers.

For removing heat after runs, there are two options. The first consists of flushing with the fluid loop. This necessitates an external reservoir of cold fluid and operation of the pump during vehicle standstill. Alternatively, and considerably simpler in usage, is module swapping. Here, two centralized modules can be swapped quickly, allowing for manual removal of spent PCM. This whole process does not rely on external hardware, and can be carried out in mere seconds. Thus, the second approach is taken.

While one PCM module is in operation, the spare is stored in a refrigerator to solidify the PCM. The spare module is expected to be refrigerated for more than 12 hours in-between series of default mode runs. With levitation mode runs less frequent, it is expected that the modules can both be refrigerated before usage, allowing for at least two full runs in a half-day. Considering this, modelling of (and designing for) the cooling process was completely neglected. It is expected that full solidification can occur on these time frames. On the Swissloop testing grounds at Innovationspark Zürich in Dübendorf, a large refrigerator (- 18 °C) can be used.

Heat Transfer Fluid

For the heat transfer between sources and PCM module, a water-based fluid will be used. Water makes for a great heat transfer fluid. Its low viscosity means minimal pumping power usage, whereas the high heat capacity makes for extraordinary heat transfer characteristics.

As an alternative to pumped fluids, heat pipes were considered. While these provide excellent low thermal resistance, they are considerably harder to flexibly integrate than a fluid coolant loop. First, their stiffness requires exact planning and tight tolerances on production, making for additional parts which need to be manufactured. Second, no commercially available heat pipes longer than 600 mm could be found. This presents problems for the application, as 0.6 m is simply too short to link all MOSFETs to the module. It also makes heat pipes unsuitable for large-scale prototypes. While not impossible, building a large evaporation-based heat transfer system was deemed out of scope for this thesis.

3.2.3 Analysis

A CFD-based transient analysis was conducted to check if a water-based coolant loop would provide sufficient heat transfer. Focus was put on the Inverter MOSFETs in default vehicle mode, since there are large power burst during these 10 second runs. For this preliminary analysis, FloEFD was used. FloEFD can be integrated with Siemens NX, and thanks to auto-meshing allows for fast simulation setup. Therefore, FloEFD was used on all fluid heat transfer simulations.

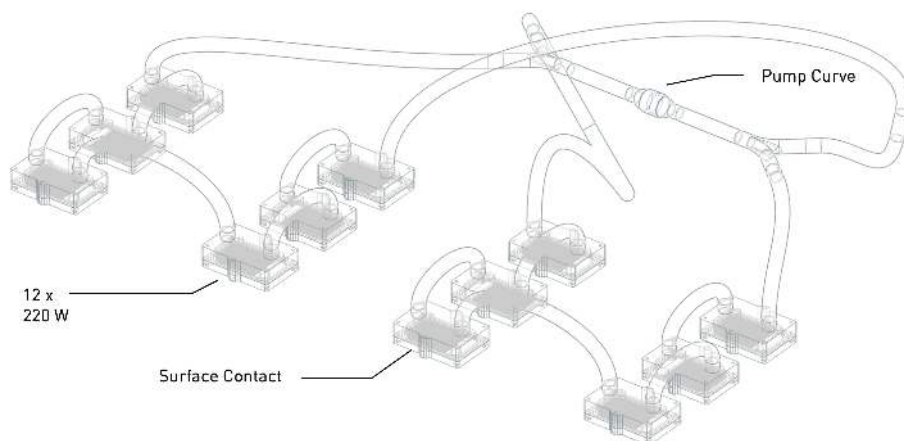


Figure 3.4: Model Setup for Initial Transient system Analysis. Two parallel arrays of six waterblocks in series were modelled with a pump curve.

Shown in Figure 3.4 is the setup used for the simulation. Here, the simulations were done without any particular mesh refinement. On the pump curve, a generic DCC pump as used in high-end PCs models the relationship between pressure loss and volume flow of the circuit. Initially, it was believed that the Inverter might use 12 MOSFETs. Here, two arrays of 6 in-series waterblocks are connected in parallel. In a vehicle, the bottom of these would be in contact with the MOSFETs. On this surface, 150

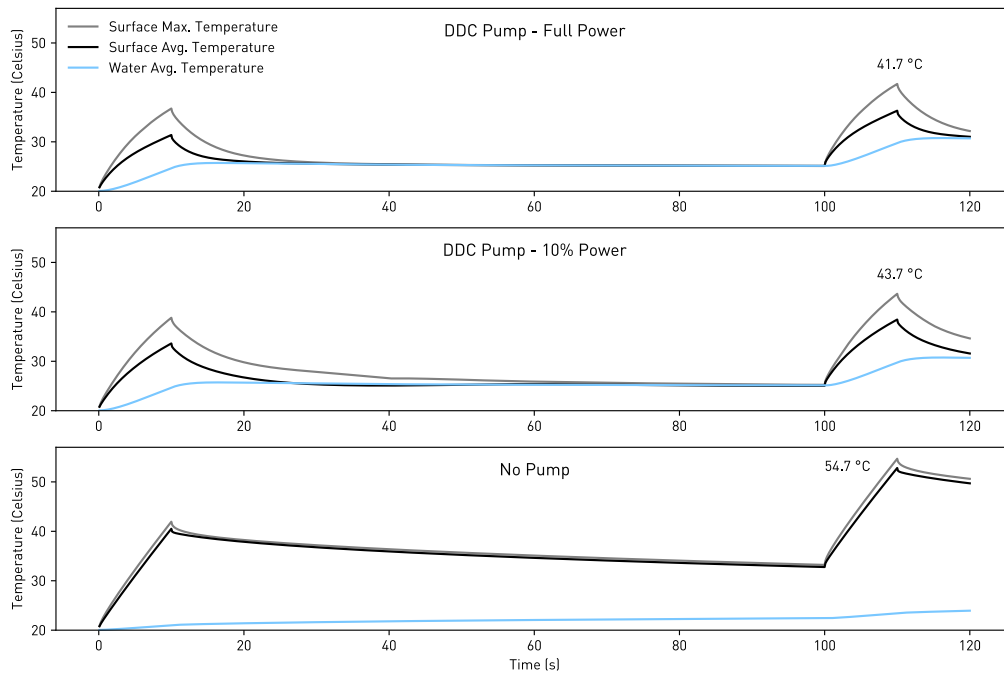


Figure 3.5: Results of Initial Transient System Analysis. If the pump is operational, the response to short power bursts is seen as adequate.

W of heat generation is applied in the simulation. The maximum and average temperature across all of these surfaces was simulated. Additionally, the average volumetric temperature of the water in the system was determined.

In order to represent the default mode, 150 W per MOSFET was applied over 10 seconds. Between these runs, a 90 second break was simulated. It must be noted that this scenario is rather conservative. Usually, the time between runs is much longer. Figure 3.5 shows the simulated temperatures for full, 10 % and 0 % pump power. The main takeaway from the simulation is that, as long as the pump is running, heat transfer is sufficient. If a centralized PCM module is added to the water loop, the time between runs can be used to transfer heat from the water to the PCM.

As expected, the last waterblock in a series of six sees higher temperatures than the first. Here, these differences have been found in the region of 0.5 - 1 °C. Notable, too, is that heat transfer with 10 % pump power is vastly better than at 0 %, but barely differs from 100 % power.

These results also allow for a limited amount of runs in full speed mode without a PCM module. To save some weight, it could be removed. With sufficient heat transfer shown to be achievable, the goal of the analysis is fulfilled.

3.3 Choice of PCM

Apart from architecture, the choice of PCM is another fundamental design decision for the prototype cooling system. As discussed in Section 2.4, a solid-liquid phase change is the most promising option for the application at hand.

3.3.1 Temperature of Phase Change

As for the temperature of phase change, the allowed minimum is roughly 37 °C. As discussed in the section defining boundary conditions for the small-scale cooling system (Section 3.1), avoiding impeded functionality at 35 °C is necessary. For the PCMs in use, the phase change does not happen at a discrete temperature. Rather, it occurs over a range of multiple degrees, centered around the temperature of phase change [19]. Thus, for hot summer days, a phase change temperature T_{pc} of at least 40 °C is mandatory. Otherwise, the PCM could start melting simply from natural convection of warm air.

There is, however, a trade-off to elevated T_{pc} . At a given heat flow, all else being equal, the temperature gradient from MOSFET to PCM must be equal. At a higher temperature of phase change, with equal temperature gradient, the electronics are hotter, too. As will be seen later, ensuring sufficient heat transfer with PCMs is a key issue. Thus, from this perspective, reducing T_{pc} is paramount to achieve sufficient heat flow.

3.3.2 Choice of PCM

With paraffin set as the PCM of choice, talks with multiple vendors were initiated. Supply chain issues also made for lackluster availability of some PCMs. For short delivery times and reduced import fees, particular care was given to choosing a supplier within the EU. The paraffin was ordered from Rubitherm Technologies GmbH (Berlin), partly thanks to their good reputation and collaboration with Hochschule Luzern (HSLU). The available Rubitherm product range for paraffin-based PCMs is shown in Table 3.1.

Table 3.1: Rubitherm Paraffin Product Range [29].

	T_{pc}	Δh_{pc}
RT 28 HC	28 °C	220 kJ/kg
RT 31	31 °C	135 kJ/kg
RT 35	35 °C	130 kJ/kg
RT 35 HC	35 °C	210 kJ/kg
RT 38	38 °C	140 kJ/kg
RT 42	42 °C	135 kJ/kg
RT 44 HC	44 °C	220 kJ/kg
RT 47	47 °C	130 kJ/kg
RT 50	50 °C	130 kJ/kg
RT 54 HC	54 °C	170 kJ/kg

As mentioned, a low T_{pc} is desirable to improve heat transfer. Meanwhile, T_{pc} needs to be sufficiently high to avoid PCM melting on warm days. To experiment with various T_{pc} , the decision was made to order both RT 35 HC and RT 44 HC. These combine suitable temperature ranges with a high enthalpy of phase change.

At $T_{pc} = 35$ °C, RT 35 HC is not suited for use on very hot summer days. It is evident that, at temperatures of 35 °C or above, partial melting occurs. However, due to the small temperature gradients of a few degrees, it is unclear if this effect is of relevance in everyday usage. Thus, real-world usage with RT 35 HC at summer ambient temperatures will be conducted. If partial melting significantly affects cooling performance, RT 44 HC will be used.

3.4 Simulation of PCM Melting

In order to design the PCM heatspreader, a procedure to model the melting process of the PCM was needed. The simulation of PCM melting processes was conducted in COMSOL 6.0. COMSOL comes with various benefits. It allows for customized physics, for example volume forces, non-linear material properties, and for large sweeps using ETH's EULER cluster.

For modelling the PCM, the "heat transfer in solids and fluids" node was coupled with the "laminar flow" node through the "ntif1" multiphysics node. The majority of the model employed here is based on Groulx et al. [30]. The effects of various modelling parameters are explained in detail in [30].

3.4.1 Modelling of Phase Change

Fundamentally, the phase change is modelled by locally varying material properties. The material is modelled as a liquid, but using very high viscosity μ below melting temperature allows for solid-like behaviour. The enthalpy of phase change is represented through a modified heat capacity.

First, a melting temperature T_m is defined over a range dT . Here, these are 35 °C and 5 K respectively. A piecewise linear function $\phi(T)$ is introduced to express the local liquid fraction of the material as shown

Table 3.2: Modelling Parameters Used in COMSOL.

Solid		Liquid		T_m	35 °C
rho_s	880 kg/m ³	rho_1	770 kg/m ³	mu_1	0.008 kg/m · s
c_ps	2 kJ/kgK	c_p1	2 kJ/kgK	beta	0.0008 1/K
k_s	0.23 W/mK	k_1	0.18 W/mK	L	210 kJ/kg
				dT	5 K
				A_mush	10 ⁶
				epsilon	10 ⁻³

in Figure 3.6. Based on this function, the modelled PCM varies its density and thermal conductivity based on the local liquid fraction. The enthalpy of phase change is represented through a modified heat capacity. A gaussian normal distribution based on the width of dT is used. In COMSOL, these material properties can be expressed analytically, meaning no interpolation between discrete values is needed. The modified heat capacity is given by [30]:

$$C_p(T) = c_{ps} + \varphi(T) \cdot (c_{p1} - c_{ps}) + L \cdot e^{-\frac{(T-T_m)^2}{(\Delta T/4)^2}} / \sqrt{\pi(\Delta T/4)^2} \quad (3.1)$$

where ΔT is equivalent to dT , L is the enthalpy of phase change, and c_{ps} and c_{p1} are the specific sensible heat capacities in solid and liquid state. In Equation 3.1, the first two terms represent the linear change in sensible heat capacity, whereas the third term represents the enthalpy of phase change. Table 3.2 shows the material properties used for the model.

The viscosity, on the other hand, is modelled using a Carman-Kozeny function [30]:

$$\mu(T) = \mu_1 + \mu_1 \cdot A_{mush} \frac{(1 - \varphi(T))^2}{(\varphi(T))^3 + \epsilon} \quad (3.2)$$

where A_{mush} influences the high viscosity in solid state, and epsilon (ϵ) is used to avoid division by 0. Effectively, this function ensures very high viscosity in the solid state and continuously decreasing viscosity with increasing liquid fraction. The viscosity is plotted in Figure 3.6. With $\rho(T)$, $k(T)$, $\mu(T)$ and $C_p(T)$ established, the Navier-Stokes equation and those for diffusive heat transfer are solved in the respective nodes in COMSOL.

To achieve bouancy, a custom volume force was used instead of the "gravity" module. This volume force was formulated as follows:

$$F_B = \rho_1 \beta g (T - T_m) \quad (3.3)$$

where β is the coefficient of thermal expansion for the liquid PCM.

Pressure Point Constraint, Meshing, Transient Solving

Due to a newer version of COMSOL being used, and some slight modifications had to be made. In particular, an additional pressure point had to be added to give a boundary condition of pressure.

Like Groulx et al. [30], the problem was simplified to a 2D model, which drastically reduced computational cost. Such a reduction allowed for rapid experimentation with various geometries and optimizing geometries using parameter sweeps to achieve adequate melt fraction at minimal weight. A very fine FE mesh was used with at a maximum 0.5 mm element size.

Liquid/Solid Fraction

To quickly assess the degree of melting in a model, the function $\varphi(T)$ was integrated over the surface area of the 2D model as follows:

$$LF(t) = \frac{1}{A} \int_0^b \int_0^h \varphi(T(x,y,t)) dx dy \quad (3.4)$$

where A is the surface of the two-dimensional model, and is used for normalization. Before any melting occurs, LF is 0. Once the PCM is fully melted, $LF = 1.0$. This variable was used extensively to assess the performance of particular geometries.

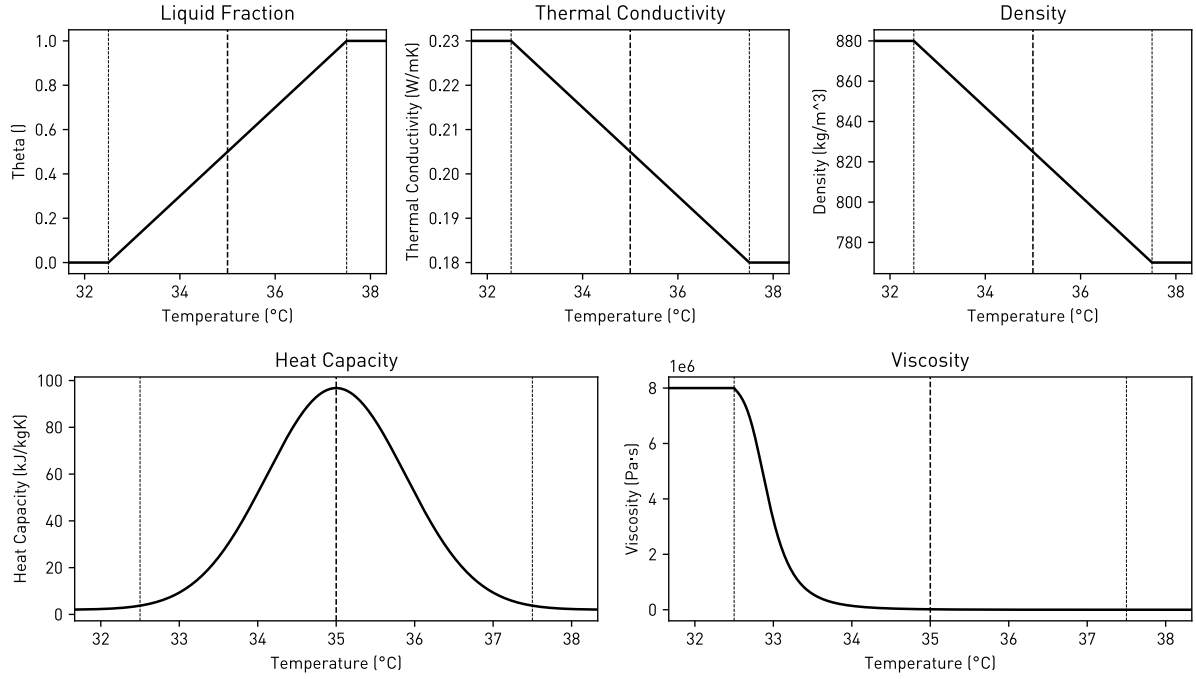


Figure 3.6: Implementation of PCM Material Behaviour for Modelling of Phase Change. The enthalpy of phase change is modelled through an adapted heat capacity. A very high viscosity at temperatures below T_{pc} is used to represent solid behaviour.

3.4.2 Convection

The results obtained by Groulx et al. [30] were successfully reproduced using the aforementioned model including natural convection. However, since solving the Navier-Stokes equations is very computationally intensive compared to solving only heat diffusion, model run times when taking convection into account were significantly higher. Partially, this was also caused by numerical instabilities, which caused simulations to diverge and fail. With the "laminar flow" and "ntif" modules enabled, very tight tolerance criteria and small time-stepping was thus required. These problems were found to be reduced significantly with "laminar flow" and "ntif" disabled. This allowed for more relaxed convergence criteria during time-stepping, and ultimately larger timesteps. For reference, if a simulation taking into account natural convection ran for 3 hours, disabling natural convection would reduce that to roughly 30 seconds.

Furthermore, it was assumed that natural convection in this case exclusively improves heat transfer. Thus, not explicitly simulating this added mode of heat transfer would represent a conservative approach. For these reasons, it was decided to not model natural convection explicitly in COMSOL. Thus, the "laminar flow" module and the multiphysics module "ntif1" which links convection and diffusion, were disabled for all following simulations.

3.5 Modelling Overview

Figure 3.7 provides an overview of the decentralized cooling system. The MOSFET's heat is convectively transferred in the waterblock (R_{wb} and Δp_{wb}). In the centralized module, there is a single water channel facilitating fluid-heatspreader heat flow (R_c and Δp_c). Both of these were modelled using FloEFD under steady-state conditions. The use of steady-state simulations allowed for extensive experimentation with various geometries and computationally costly optimizations. At the heart of designing the waterblock and module water channel is the trade-off between pressure loss and heat transfer. These are explored in detail in Sections 3.6.2 and 3.7.

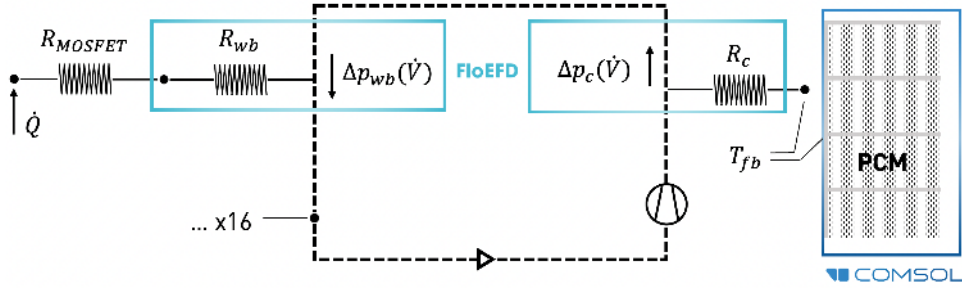


Figure 3.7: Overview of Simulation Strategy. To optimize waterblock and the module's water channel, FloEFD was used. Phase change simulations were conducted in COMSOL in 2D.

The PCM was modelled together with a heatspreader (fin) array in COMSOL. To simplify the interface between the two simulation tools, a fin base temperature T_{fb} of 50 °C was assumed. At 15 °C over the melting temperature of RT 35 HC, good heat transfer is obtainable in the module. A fin base temperature T_{fb} of 50 °C is a conservative working point, since it allows for large temperature differences of roughly 100 °C between the fin base and MOSFET junctions. Fixing T_{fb} early on allowed for simultaneous work in COMSOL and FloEFD, which eased the design process. Towards the end of this chapter, all thermal resistances and pressure losses will be known. A short overview and assessment of system performance will be given.

3.6 Design: PCM Module

The centralized PCM module is fabricated using Selective Laser Sintering (SLS) with an aluminum alloy, AlSi10Mg [31]. While this manufacturing technique is not suited for mass production, single parts can be produced without a cast, mold or similar. Additionally, it enables complex geometries which would not be easily achievable with milling. Lastly, SLS enables the manufacturing of lattice infills, which makes for very strong and light parts. On a lightweight prototype, SLS is thus the perfect manufacturing process for a prototype multi-chamber PCM container.

The maximum printing volume available is 245 mm x 245 mm with 380 mm printing height. The manufacturing with SLS dictated maximum overhang of 45 °, since otherwise solid metal supports are needed. These cause additional machine time and manual labor. The minimum vertical wall thickness is 0.5 mm. With SLS printing established as the manufacturing process, various heatspreader types were considered.

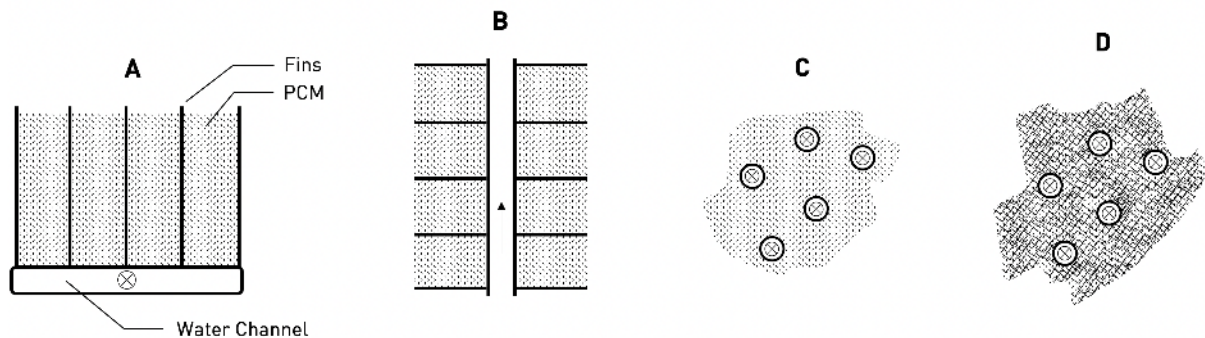


Figure 3.8: Heat Spreader Concepts. Options A and B are traditional channel-and-fin approaches. Concept C and D represent distributed channel approaches which are only attainable using additive manufacturing. Option D uses a fine lattice to enhance heat transfer into the PCM.

Some of the approaches considered are shown in Figure 3.8. Ideas A and B represent more traditional approaches, with straight water channels and ordinary fins. Idea B could also have been implemented in a cylindrical format, but this idea was not pursued further due to the difficulty of printing large overhanging structures using SLS. As an alternative to A and B, option C consists of small water channels

distributed through the PCM. Option D is similar, but with an added lattice structure to facilitate heat conduction across the PCM. Option C and D represent exciting options, which could have been uniquely manufactured thanks to the SLS process. A generative design method could have been used to find the optimal lattice and water channel structures. However, these ideas were eventually discarded for two reasons. First, between modelling the PCM, designing the centralized module and additional waterblocks for the MOSFETs, little time was left to experiment with extravagant techniques. Second, it was assumed that an approach like D only unfolds its full potential when optimized in 3D. Due to the high computational cost of simulating PCM's in 3D, this was not seen as feasible.

After extensive experimentation in COMSOL, configuration B was shown to be a more effective layout than A. Thus, option B was pursued in detail.

3.6.1 Optimization of PCM Heatspreader

The heatspreader according to concept B was modelled and optimized according to the techniques discussed in Section 3.4. Here, the goal was to ensure adequate heat transfer while being as lightweight as possible. Good wetting between heatspreader and PCM was assumed, so that the contact resistance between aluminium and PCM could be neglected. With constant fin base temperature T_{fb} , the model was kept as simple as possible. Half a fin and half a PCM pocket was simulated.

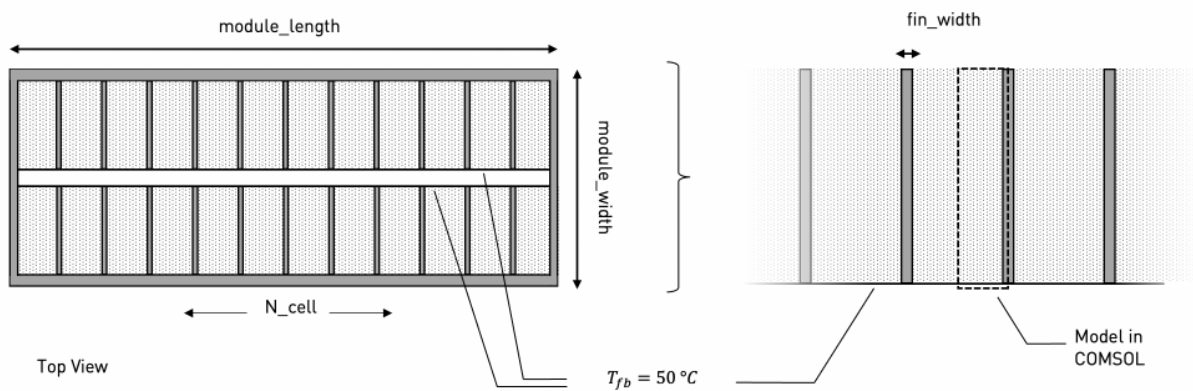


Figure 3.9: Phase Change Modelling Setup. The model employed in COMSOL as shown on the right was chosen to be the smallest unit cell to reduce computational cost.

As discussed, the design of the module was constrained by the build volume of the manufacturing process. This was taken into account in the simulations. A crude formula to assess mass based on a constant container wall thickness, fin count, and fin thickness was established. After structural refinement, the mass of the PCM module could be significantly reduced by using lattices, local stiffening elements and thin walls. However, the straightforward mass approximation performed well to compare various configurations. The volume of PCM was computed to satisfy a capacity of 440 W over 15 minutes, which conservatively conforms with the critical levitation mode. Sufficient heat transfer at the given capacity and T_{fb} was ensured by assessing the melt fraction of the PCM after 15 minutes. This value is based on Equation 3.4. The model setup is shown in Figure 3.9. In the model, a constant container wall thickness of 2 mm was assumed. The fin width, module width, module height, and number of cells N_{cell_x} were implemented as variable parameters. The module length is equal to 200 mm and given by the build volume of 245 mm subtracted by 2x 20 mm for lattice reinforced handles. These are shown later. Figure 3.9 shows the top view of the model. This setup is suited for approach B with a singular water channel which runs through the middle of the module. It allows for all simulations to be reduced to 2D. Natural convection can not be accounted for from this perspective with a 2D model. This is not a major disadvantage, however, since simulating convection was seen as infeasible due to its computational cost anyway. Only heat diffusion was simulated.

The module length was predetermined to conform with maximal print dimensions. The module height was defined dynamically based on the module width, length, and PCM volume. The PCM volume, in turn, was determined based on the capacity goal of 440 W over 15 minutes. Overall, this reduces the independent variables to fin_width , $module_width$, and N_{cell_x} .

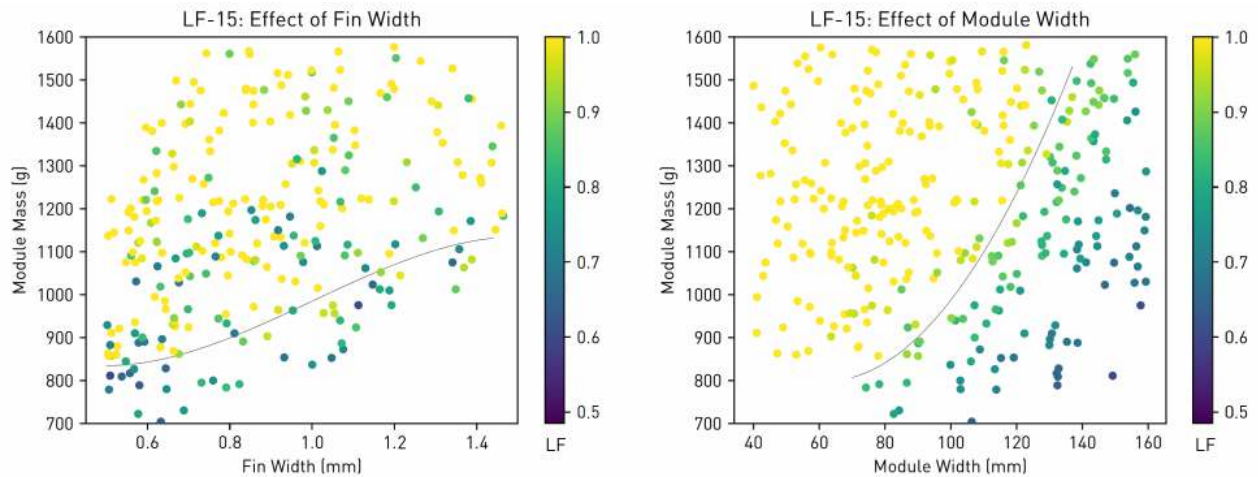


Figure 3.10: Effect of Fin Width and Module Width on Melt Fraction and Module Mass. The contours in black show the front of acceptable solutions with adequate melt fraction. Reducing fin and module width are weight-effective design strategies.

The effect of these three variables was examined in detail. Figure 3.10 shows the findings. Here, COMSOL's Monte-Carlo search algorithm module was used to minimize for mass. The Monte-Carlo module can explore given design space based on random point sampling.

As stated, the requirement for a minimum liquid fraction at 15 minutes (LF-15) of 95 % has been set. Since heat transfer correlates with heatspreader mass, less mass will make for lower LF-15 values. It is thus evident that an algorithm minimizing mass while sticking to this criterion will come to optimal solutions which are very close to the criterion. For example, it is unlikely that the best result will be a heatspreader which achieves full melting after 5 minutes and contains excess mass. The black lines in Figure 3.10 roughly mark the zone of $LF-15 = 0.95$. Both of these lines show that, for adequate heat transfer, mass can be reduced by reducing both fin width and module width. The relationship of mass and module width can intuitively be explained by the fact that heat transfer benefits from a short thermal path. This path is reduced if the average distance between PCM and the fin base is reduced.

For the number of cells, no clear correlation has been found. Thus, a design was fixed as follows. First, the maximum module length and height were determined. The module length is limited by the print volume. The maximal module height is given by the outer fairing of the vehicle. Over vehicle iterations, this height changed. Thus, the process of finding an optimal design had to be repeated multiple times. Both the module height and length were maximized until limiting factors were encountered. By doing so, the module width could be kept minimal, which benefits weight as discussed above. Due to these constraints, a module width of 94 mm could be achieved. Next, the fin width was determined. Since narrow fins benefit weight as shown in Figure 3.10, this parameter was minimized. A fin width of 0.5 mm was ultimately chosen. This is the narrowest wall thickness the supplier was willing to manufacture. Lastly, as the only remaining parameter, N_{cell_x} was varied to achieve sufficient heat transfer. With 0.5 mm fins, the minimum value of N_{cell_x} to satisfy a melt fraction LF-15 of 0.95 was found to be 39. This makes for a total of 38 fins. Figure 3.11 shows the melting process of this final, optimized geometry. A single cell with two fin halves to either side is shown. With an overall LF of 95.3 %, 400 W can easily be absorbed over 15 min. This is because the overall capacity is sufficient for 440 W. It is expected that in a non-idealized scenario, the melting will be accelerated by conduction in the container walls.

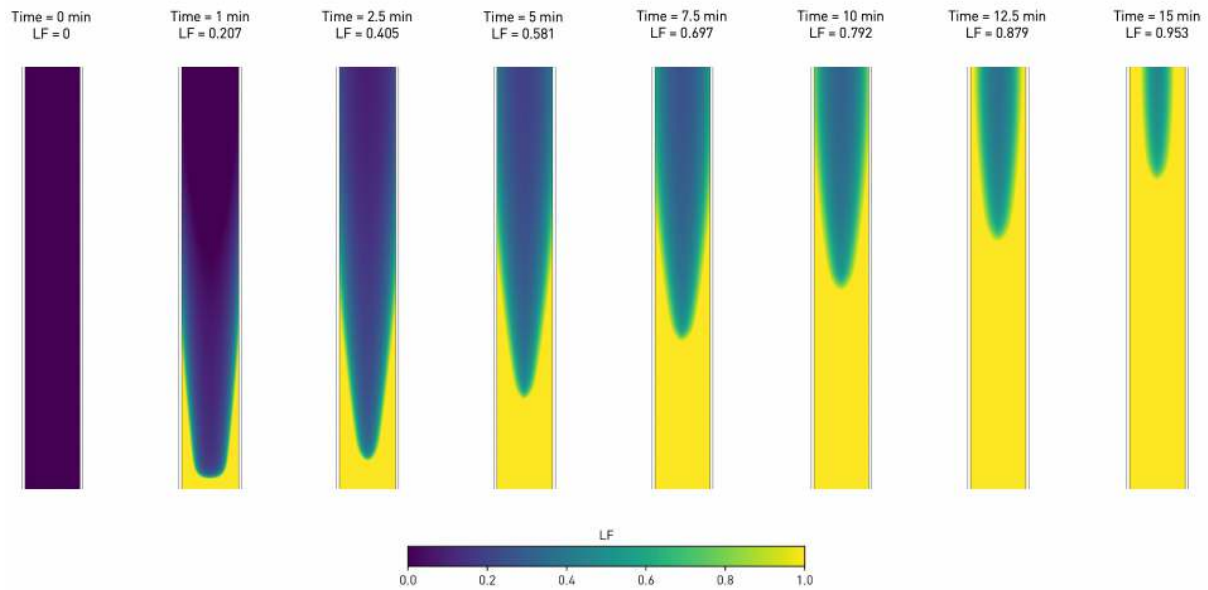


Figure 3.11: Visualization of PCM Melting Process in Optimal Heatspreader Design. Here, the local melt fraction is shown in color. The average melt fraction is denoted at various time steps, reaching 95.3 % at 15 minutes. The shown cells correspond to the sections modelled in COMSOL as shown in Figure 3.9.

3.6.2 Optimization of Water Channel

A water channel was integrated into the design to facilitate the heat transfer between water and PCM. Here, sufficient heat transfer to avoid a thermal bottleneck was a high priority. Additionally, the pressure loss associated with the coolant flow should be kept adequately low to enable 180 L/h of throughput in the coolant loop. To obtain good heat transfer, placing a lattice in the water channel was considered. However, it was found that CFD modelling of such a lattice using FloEFD is practically infeasible due to the computational resources consumed. Placing fins in the channel was also considered, but the printing process did not allow for horizontal fins to be printed. Thus, as a means of improving heat transfer without adding surface area, the channel's design leveraged turbulence. Generally, turbulence makes for mixing of the heat transfer fluid, which allows for a larger portion of the flow to be exposed to the heat transfer surfaces. Two means of creating increased turbulence were identified. First, when decreasing the channel width, higher velocities are reached for equal volume throughput. The associated increase in velocity gradients in the boundary layer cause increased vorticity. Second, small teeth were added to the channel walls. These create local flow detachment, resulting in increased turbulence. The teeth are visible in more detail in Figure 3.14.

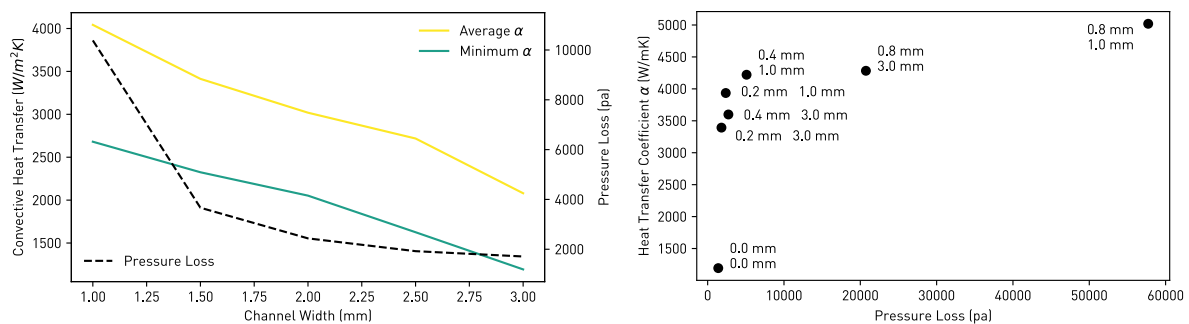


Figure 3.12: Water Channel Heat Transfer and Pressure Loss Simulation Results. Left: Heat transfer and pressure loss dependence on water channel width. For this simulation, the teeth were fixed at 3 mm long and 0.2 mm high. Right: Heat transfer and pressure loss at various teeth geometries. Here, the channel width was fixed to 2 mm.

To assess and optimize these techniques, a FloEFD steady-state model was set up. To simulate the heat transfer, water entering the module at 60 °C was assumed. A uniform 440 W heat sink on the fin base was used to simulate the heat absorbed by the PCM-fin array. This represents the critical levitation scenario.

Figure 3.12 shows the results of these simulations. The heat transfer coefficient shown was determined by comparing average and minimal fin base temperature with the in-flowing water temperature. The decision was made to opt for a channel width of 2 mm and teeth dimensions of 0.2 mm / 3 mm. While heat transfer could have been improved significantly using a smaller channel, the chosen dimensions comfortably ensured sufficient heat transfer. At 180 L/h, the pressure loss over the water channel is simulated to be 1600 Pa. In FloEFD, the pressure loss was assessed by subtracting the fluid exit port pressure from the entrance port pressure. The temperature difference between the in-flowing water and fin base ΔT_c is 3.2 °C on average. The temperature distribution is not perfectly homogeneous along the finbase as can be seen in Figure 3.13. However, these differences were neglected for the sake of simplicity. At larger channel widths, over 3 mm, unequal flow velocities along the channel height were created due to the vertical momentum of the incoming water flow. The inhomogeneous flow patterns caused high variations in finbase temperature. At the shown 2 mm channel width, the flow velocities along channel height were adequately homogeneous.

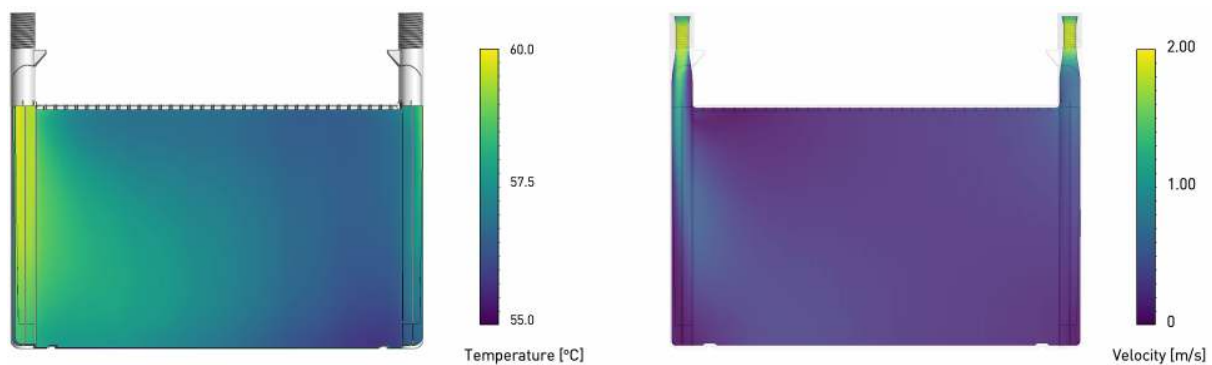


Figure 3.13: Water Channel Temperature and Flow Velocity Simulations. Left: Finbase temperature distribution at 60 °C water in-flow. Right: Homogeneous velocity distribution in water channel.

3.6.3 Structural Design

Structurally, the single largest load occurring on the module originates from internal pressure. This pressure stems from two sources. First, if the cooling system is operating in vacuum, an effective over pressure of 1 bar is experienced in the inside chambers. Second, if the PCM is in its fully melted state, it expands. This expansion causes the air in the PCM chamber to compress, thus causing excess pressure. Since the chambers are sealed to avoid leakage, the module needs to be designed to accommodate the resulting stresses.

Briefly, the idea of incorporating pressure relief valves into the lid was considered. One valve would ensure that the inside pressures over 1 bar would not be exceeded. However, with such a valve, PCM cool-down would cause an under-pressure in the module, even more so after use in near-vacuum. Sub-ambient pressure would cause the fins to be loaded in compression, making for challenging buckling simulations. Thus, another valve would have been added to avoid low pressures. This solution employing two valves was discarded, mostly due to its complexity and the significant weight the valves would introduce.

In addition to internal stresses, the module is designed to accommodate other minor forms of local loads. Small reinforcements and edge blends are employed around a mounting hardware interface attaching module to vehicle. (9. in Figure 3.16. At the time of writing, the mounting hardware has not been produced. It will consist of clips, which are mounted to the vehicles chassis using blind rivet nuts. These clips are based on an existing design by Swissloop which can be easily opened using high voltage gloves. They will be made using fused deposition modelling (FDM) 3D-printing.

Airflow and Excess Pressure

Based on the ideal gas law, the pressure p_m inside the module after full melting is given by:

$$\frac{p_m}{p_s} = \frac{V_{air,s}}{V_{air,s} - \Delta V_{pcm}} \quad (3.5)$$

with p_s being the pressure when the PCM is solid, $V_{air,s}$ the overall volume of air within the module if the PCM is solid, and ΔV_{pcm} being the volume change of the PCM upon melting. Here, p_m is set to 1 bar, since the module is sealed when the PCM is cold. Due to seal imperfections, it is assumed that over long periods of time $p_s = 1$ bar in the module is unavoidable.

To reduce the structural load caused by internal pressure, p_m should be reduced. The module width, length and PCM fill height are given. These have been determined earlier to conform with capacity goals, maximal print dimensions, and heat transfer criteria. Thus, V_{pcm} was considered given. With $V_{air,s}$ as the only tunable design parameter to influence p_m , some steps were taken to increase the internal air volume of the module. As a first step, the air gap between PCM and lid in the chamber was increased to 26 mm. Second, the air volume within the body and lid lattice infills is connected through holes to the PCM chamber. In effect, these holes enable free air flow between the air gap in the PCM chamber and the infill air volumes, significantly increasing $V_{air,s}$. With these steps, p_m is expected to be 1.76 bar.

Structural Design

Despite p_m calculated to be 1.76 bar, the module is designed to tolerate 3 bar. This is for two reasons. First, a catastrophic structural failure would be accompanied by air and PCM rapidly exiting the module. The kinetic energy could cause damage to bystanders. Second, the exiting paraffin could significantly damage other vehicle components, rendering custom-made parts useless. Hence, it was opted to allow for more pressure.

Structurally, the module consists of two main zones. In the lower part of the body, the fins (11. in Figure 3.14) directly transfer the stresses induced by internal pressure between the module walls. Here, the body's outer walls are relatively thin (1.5 mm), since they are directly supported. Meanwhile, in the upper part of the body where no fins are present, the walls are stiffened using a sandwich construction. This is achieved using a lattice, which connects the 1.5 mm wall to a 1 mm outer face sheet. Between the section with stiffened walls and that supported by fins, there is a 10 mm overlap to reduce stress concentrations. The core (body lattice, 7. in Figure 3.14) thickness is 10 mm along the long edge and 20 mm at the short edge of the body.

Like the body's upper part, the lid is constructed using 1.5 mm face sheets and a 6 mm core. At below 400 g, it is significantly lighter than an equivalently strong CNC milled part. The lid lattice core (3. in Figure 3.14) is reinforced by a solid lip around the lid's edges. This solid frame takes up screw compression and stiffens the gasket interface. Similarly, the body is equipped with stiffeners (6. in Figure 3.14). At a height of 6 mm, these significantly increase the rigidity of the body's gasket surface. This, in turn, decreases the likelihood of leaks upon loading with higher pressures.

The lattice structure chosen for the body and lid is "dode" with a unit length of 5 mm, and approximately 15 % volumetric infill. It was chosen since the manufacturer (Feramic AG, Stallikon) has extensive experience with this lattice type. On the lid, the air entrance hole as shown under 4. in Figure 3.14 contains an M6 thread. On this thread, a GORE PolyVent XS plug [32] is mounted. It is fitted with a membrane, allowing air to freely pass while blocking the flow of paraffin. This avoids contamination of the lid core with PCM during rapid accelerations and decelerations when levitating.

FEM Structural Analysis

Finite Element Method (FEM) structural analysis was performed on both lid and body using Siemens NX Nastran. Modelling the lattice posed a significant problem. Rendering the fine lattice structures using 3D or 1D elements was determined to be very computationally expensive for the size of the model at hand. Thus, the lattice cores had to be modelled as an equivalent solid body, as seen in 3.16. Despite ongoing research into predicting equivalent material characteristics of lattice structures [33], there are no easily obtainable values for any lattice type. Some lattices have been characterized in detail (e.g. the octet truss lattice [34]), but with a vast range of lattice types many have not been investigated in detail. Additionally, it is presumed that failure in these lattices is a local phenomenon and is highly dependent on manufacturing defects and load application.

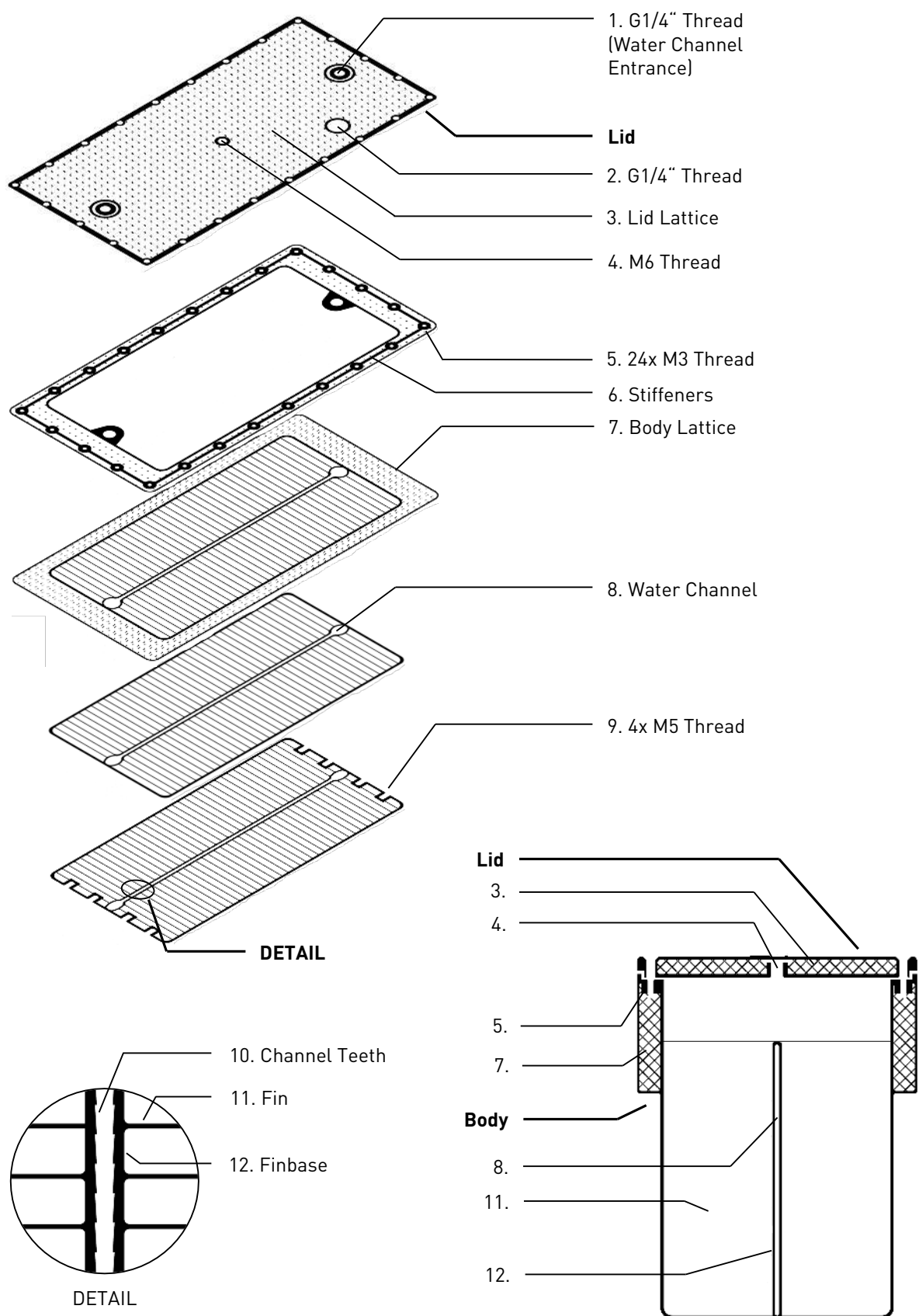


Figure 3.14: Module Cross-Sections. Overview of Key Design Features.

Therefore, the lattice was assumed to be a solid material with isotropic material properties, and the strength and stiffness were experimented with. Figure 3.16 show a lattice with 10 % the stiffness and 5 % the strength of solid the aluminium in the face sheets. Generally, it was found that the lattice is not subject to stresses above 5 Mpa. At 5 % strength of 220 Mpa, 5 Mpa is equivalent to a safety factor S_f of 2.2. The lid was modelled using 2D shell elements. The result of 0.3 Mpa (3 bar) of pressure with fixed constraints at the screw holes can be found in Figure 3.15. The stresses in the face sheets should not exceed 74 Mpa, making for a safety factor S_f of 3.

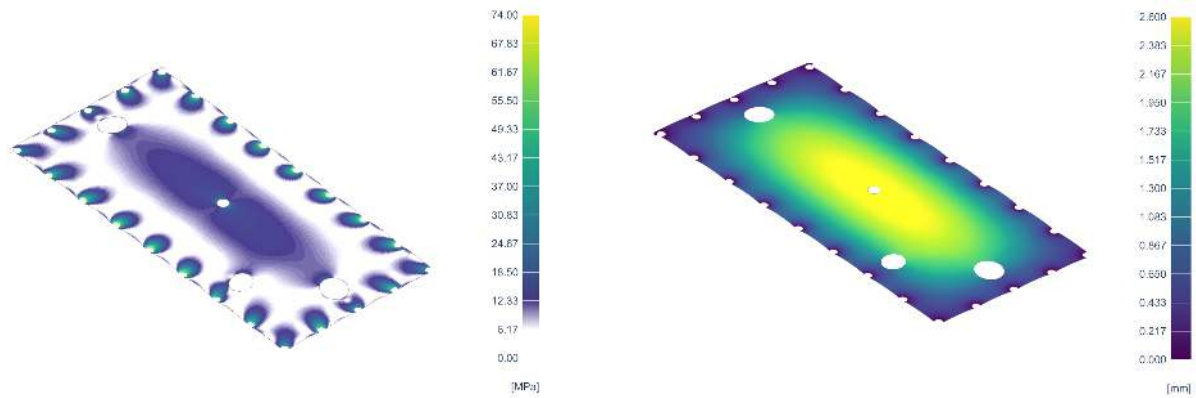


Figure 3.15: FEM Results of Structural Simulations for Module Lid. Left: Stresses in lid facesheet at 3 bar relative internal pressure. Right: Deformation of lid.

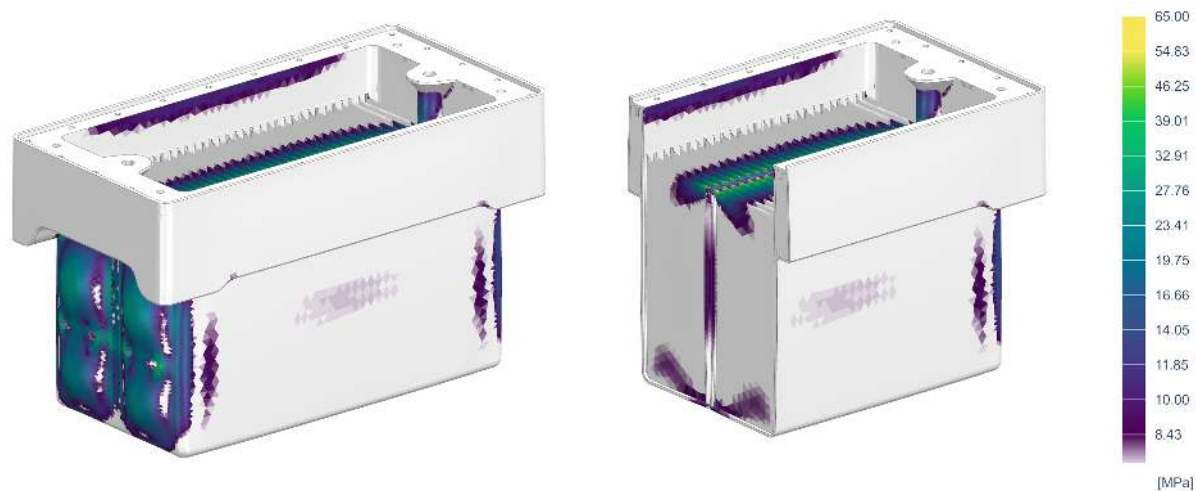


Figure 3.16: FEM Results of Structural Simulations for Module Body. Shown are Von-Mises stresses in module body at 3 bar relative internal pressure.

The module body was modelled using 3D tetragonal elements. 0.3 Mpa of pressure was applied on all outer surfaces. A fixed constraint on the whole gasket sealing area was applied, meaning the effectiveness of the stiffeners was not modelled. Significant stress concentrations were found in the fins and outer wall. Edge blends were added to slightly decrease these peaks. Still, with maximum Von-Mises stresses below 65 Mpa, a safety factor of 3.3 could be reached. The resulting stresses are shown in Figure 3.16.

3.6.4 Module Overview

Shown in Table 3.3 is a weight overview of the module. Note how the ratio of PCM to metal mass is roughly 50 %. On the body, the weight is distributed roughly equally between heat transfer devices (fins, water channels) and structural features (walls, lattice and air pockets).

Table 3.3: Module Weight Overview

Fins	542 g
Water channel (excl. water)	201 g
Body structural walls	726 g
Body lattice	139 g
Body	1469 g
Lid lattice	57 g
Lid structural walls	291 g
Lid	348 g
Screws	19 g
PolyVent XS Plug	5 g
Gasket	12 g
PCM	1727 g
Module Total	3580 g

Figures 3.17 and 3.18 show the finished module shortly before PCM fill-in. On the lid, some text was added for version and material identification. In Figure 3.18, the fluid couplings and the pressure sensor are visibly mounted. The membrane plug is not visible, since it is located on the lid's surface facing inside. The lattice 20 mm at the ends to make for an integrated handle. Testing with mock-ups has been conducted to make sure that the module can be lifted using high voltage gloves.



Figure 3.17: Finished Module Body without Lid, Gasket and PCM.



Figure 3.18: Finished Module with Lid, Pressure Sensor, and Quick Connectors.

3.7 Design: Waterblocks

The waterblocks in the decentralized cooling system transfer heat from MOSFETs into the coolant loop. They consist of three major components: Lid, gasket and base. The metal base transfers the heat from the MOSFET to the transfer medium. Early on, it was decided that the base should be CNC milled instead of SLS printed. Due to the amount needed (17 on the vehicle, 3 as spare), SLS printing was out of the question due to sponsoring and cost constraints.

The material of choice is aluminium ($\lambda = 140 \text{ W/mK}$ [35]). While copper has better thermal conductivity, it is denser by a factor of three and thus heavier, harder to obtain, and more difficult to mill.

To maximize convective heat transfer, the surface contact is designed to be large. This is achieved by arranging a large number of narrow fins in the flow path. To enable narrow channels between fins, a 0.6 mm mill head was employed. This is the smallest size available by the manufacturer. With a mill head diameter of 0.6 mm, the minimal achievable cut width is 0.6 mm. To avoid excessive vibrations during milling, the cut depth should not exceed five times the mill head diameter. Therefore, the maximum cut width is 3 mm. Additionally, all design features of the waterblock base have to be accessible with a five-axis CNC machine.

Optimization of Waterblocks

A simplified, parametric model of the waterblock was built in Siemens NX. Two design metrics were set to quantify the quality of a design. The pressure loss quantifies the resistance of a waterblock. Since there are 17 waterblocks in series, the flow resistance should be kept within bounds to enable adequate flow. The pressure loss and volume flow are discussed in detail in Section 3.10.1. With conduction through the base nearly negligible, the second key metric is convective heat transfer. It is quantified by measuring the average temperature on the surface contacting the MOSFET at a given heat flux. The simulations for design optimizations were performed at steady state, with an inflow of $\dot{V} = 180 \text{ L/h}$. The temperature of the incoming water was set at 40°C , with outflow temperatures unconstrained. A constant heat flux of 150 W was applied at the MOSFET contact surface. The simulations were performed on only a single waterblock, saving computational time and allowing for more design variation.

As shown in Figure 3.19, the flow entrance is centered in the fin array, meaning the flow passes outwards in both directions. The fin array is slightly offset to make space for the flow exit. The waterblock's outer dimensions were given by space constraint on the Inverter PCB.

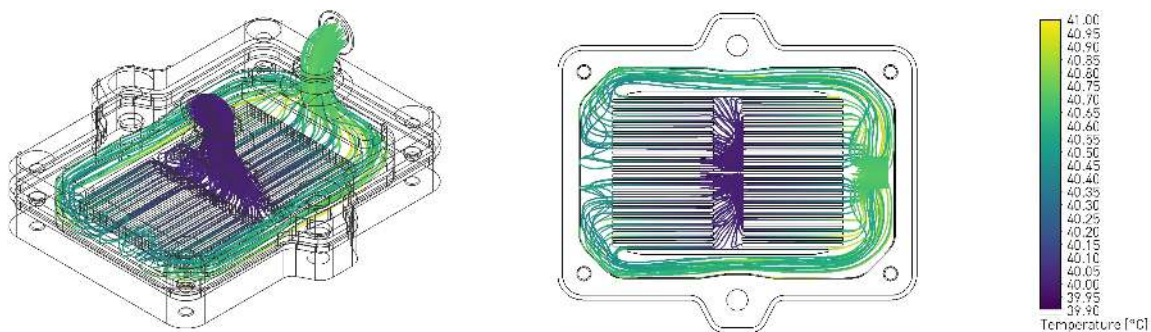


Figure 3.19: Simulated Flow Temperature in Waterblock. As in the final design, the water enters centrally at low temperatures. After passing through the fins outwards, it is collected at a single outwards-facing outlet.

Some design parameters were not studied in detail. For example, more time could have been spent on investigating the advantages of using a fin array twice as long with unidirectional flow. In this specific scenario, such a left-to-right flow layout had to be discarded due to space constraints. With fittings standing off the lid significantly in two directions instead of one, PCB space between MOSFETs would have had to be increased.

To find the optimal design for the model at hand, a multiparameter optimization was performed in FloEFD. The optimization mode, HEEDS (Hierarchical Evolutionary Engineering Design System), employs search algorithms to find optimal solutions across a design space of possible geometries. Here, the number of variable parameters was limited to four to limit computational cost. The fin (array) length,

cut depth (equivalent to fin height), fin array width and cut width could all be varied. The cut width is equivalent to the distance between two fins. The fin width, which was not varied, is set equal to the cut width. For all of these, minimal and maximal boundaries were set based on manufacturing or space constraints.

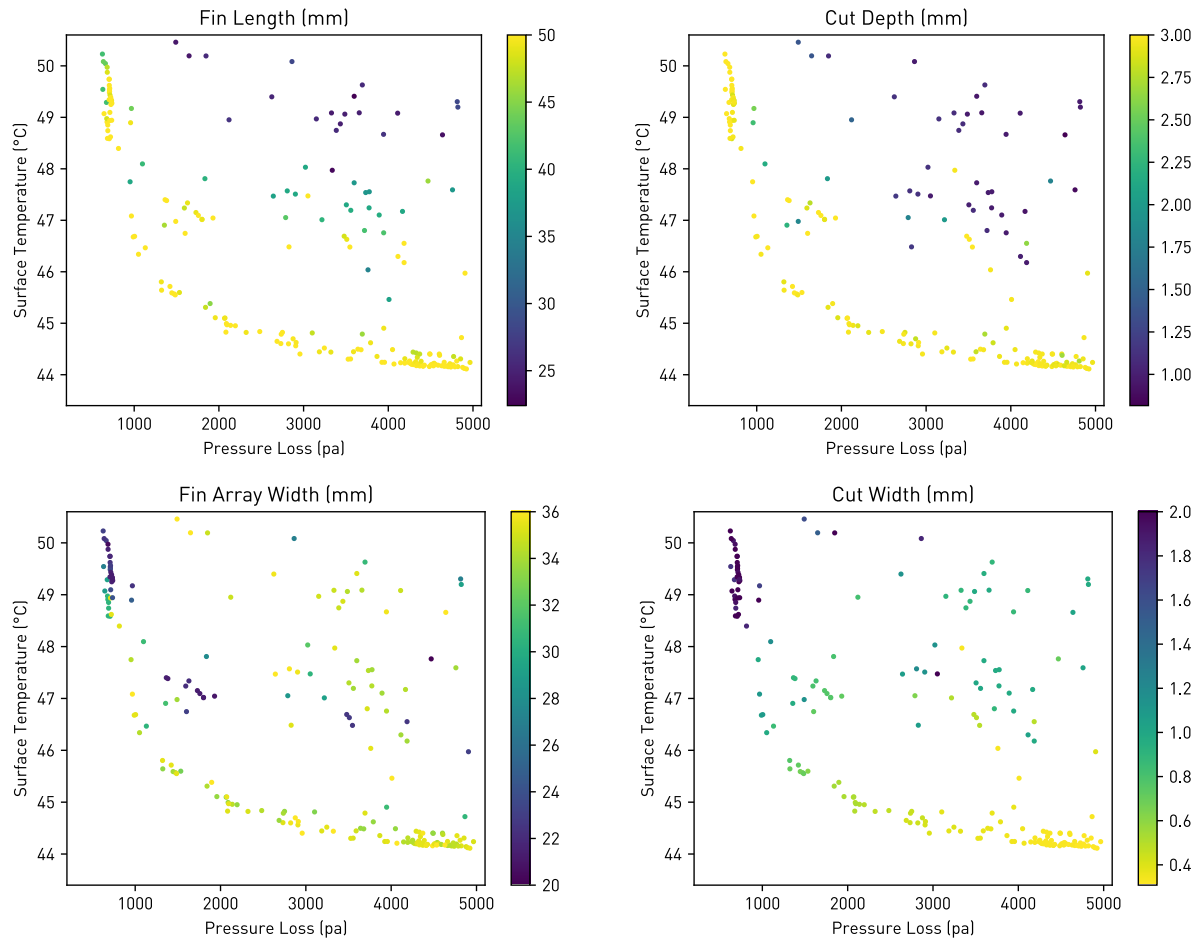


Figure 3.20: Effect of Waterblock Design Parameters on Heat Transfer and Pressure Loss. The colors indicate parameter dimensions.

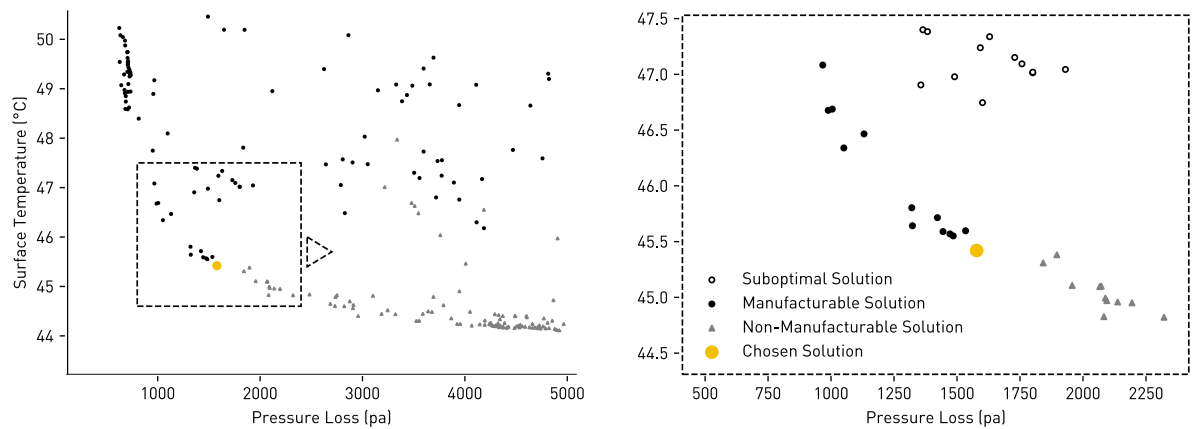


Figure 3.21: Choice of Optimized, Manufacturable Waterblock Design. The chosen design displays lowest MOSFET surface temperatures amongst all manufacturable solutions.

Shown in Figure 3.20 are the results of the optimization. A clear pareto front is visible, depicting the set of nondominated solutions. For all four geometric parameters, clear trends are visible. For both the fin length and fin height (cut depth), more is better. The fins should be as long as possible and as high as possible, increasing overall surface area. For the fin array width and cut width, on the other hand, the ideal dimensions depend on the required heat transfer and pressure loss. For low heat transfer with low pressure loss, a narrow fin array with wide cuts is ideal. Conversely, for increased heat transfer with high pressure loss a wide fin array with narrow cuts is ideal.

Shown in Figure 3.21 is the same set of results as above. On the pareto front, the optimal design depends on the prioritization of pressure loss and heat transfer. No analytical trade-off formulation was derived here. Rather, the pareto solution with the best heat transfer and adequate manufacturability was chosen. This solution also featured reasonable pressure loss.

The final design features a fin length of 50 mm, cut depth of 3 mm, a fin array width of nearly 36 mm and a cut width of 0.6 mm. With that, it makes full use of the available manufacturing capabilities at hand. At 180 L/h, the pressure loss of the optimized design is 1580 Pa. At 150 W, the temperature of the base surface in contact with the MOSFET is 5.42 °C over the water temperature at the inlet.

Characterization

In case the waterblocks will be used again in future Swissloop vehicles, the correlations for pressure and surface temperature as a function of volume flow were simulated. To find adequate relationships to fit the simulated operating points to, it was first determined whether the flow is laminar or turbulent. With the given number of fins, the flow of 180 L/h has to pass through approximately 1 cm² of overall area between the fins, making for average flow velocities of 0.5 m/s. According to Equation 2.3, $Re < 10^5$. Therefore, the flow is assumed to be predominantly laminar inside the waterblocks. For laminar flow, the Nusselt number according to Equation 2.9 is proportional to $\sqrt{u_\infty}$, and thus:

$$T_{surface} - T_{water} \propto 1/\overline{Nu} \propto 1/\sqrt{u_\infty} \propto 1/\sqrt{\dot{v}}. \quad (3.6)$$

Accordingly, for the MOSFET surface temperature, the correlation was formulated as follows:

$$T(\dot{v}) = a + b \cdot \frac{1}{\sqrt{\dot{v}}}. \quad (3.7)$$

A linear least squares fit found that $a = 41.22$ °C, and $b = 56.73$ °C $\sqrt{L/h}$. Similarly, the pressure loss follows the following correlation [36]:

$$\Delta p = \zeta \cdot \rho \frac{\bar{u}^2}{2} \quad (3.8)$$

where ζ is a dimensionless coefficient. From this follows that $\Delta p \propto \dot{v}^2$. Thus, the correlation between pressure loss and volume flow is described as:

$$\Delta p(\dot{v}) = c \cdot \dot{v}^2. \quad (3.9)$$

A linear least squares fit found $c = 0.041 \text{ Pa} \frac{h^2}{L^2}$. Shown in Figure 3.22 are the results at varying volume flow $\dot{v}$. The correlations only take into account the data points under 500 L/h, since operation above that is very unlikely given similar pump hardware.

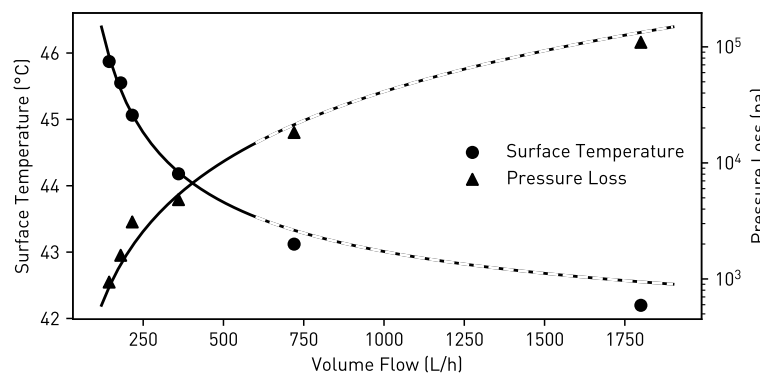


Figure 3.22: Waterblock Pressure Loss and Temperature Dependence on Volume Flow.

Waterblock Overview

After milling, three waterblocks were anodized to avoid corrosion during long-term operation. Small sections were spared to allow for adequate grounding. The electrical grounding of metal heatsinks is needed to minimize emission of EMI (electromagnetic interference), of which the MOSFETs are the vehicle's main emitter. The waterblocks will be grounded using a small washer cable.

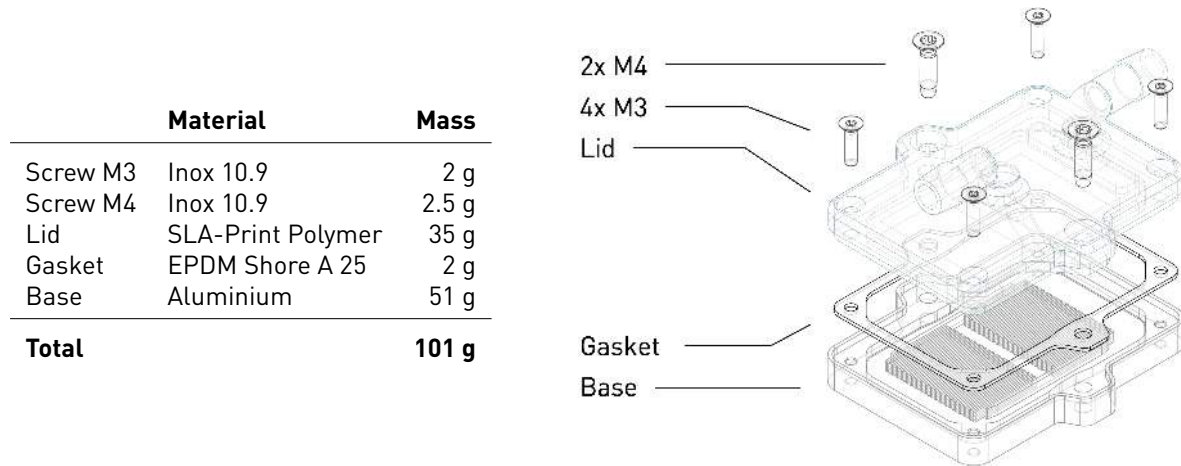


Figure 3.23: Waterblock Design Overview.

A gasket is used as it has shown to be significantly more space-effective than an O-ring, which needs a dedicated, wide groove with large radii at corners.

The lid's fluid entrance and exit are specifically arranged to minimize the allowable distance between MOSFETs. Two longer M4 screws are used to clamp the whole waterblock onto the PCB, compressing the the whole assembly against the MOSFET for adequate thermal contact. The MOSFETs come with ready-to-use thermal paste, which is used here. Figure 3.23 provides a mass and design overview.

3.8 Integration

Below, some of the accessories are covered in detail. These are not relevant to the core findings of the thesis, but still need to be covered to make for a functional thermal management system.

3.8.1 Pump

For the pump, a standard PC water cooling pump was used. These models are built for long-term usage, are lightweight and compact. While brands vary, most designs follow two major pump types. The D5 models are suited for slightly higher volume throughput, but produce less head pressures than DCC models. The D5 is able to cool itself through the pumped medium, whereas the DCC pump is reliant on a heatsink for operation over longer timeframes. Commercially available D5 pumps are roughly 100 g heavier than their DCC counterparts (without heatsink).

For the application at hand, the high number of waterblocks means that volume throughput is low and head pressure needs to be high. Thus, the decision was made to choose a DCC pump. A target of 180 L/h volume flow was set early on, with which simulations for the waterblock base and module channel were conducted. in the vehicle, the pump will be powered directly by the Vehicle Control Unit (VCU), which was outfitted with an onboard MOSFET capable of 12 V. It will be seen during vehicle testing during summer if a heatsink is necessary. Table 3.4 provides a detailed overview of pump weights in different configurations. The aluminium casing serves as a heatsink.

3.8.2 Quick Connectors

To quickly disconnect the module from the coolant circuit, drip-free couplings are used. The SPT05.7101 / L/CG and SPT05.1101 / L/CG from Stäubli AG are very lightweight due to their full aluminium construction. The couplings are shown in Figure 3.24. It was expected that, using these, no air can enter the

	Pump Assembly 1	Pump Assembly 2
Pump + Cables	152 g	152 g
Aluminium Casing	132 g	-
Plastic Casing	-	20 g
Transparent Lid	76 g	-
Plastic Lid	-	39 g
Screws	24 g	24 g
Total	384 g	235 g

Table 3.4: Pump Configuration Overview.

coolant loop. On the vehicle side of the couplings, 90 ° fittings are used. These allow for good grip with high voltage gloves and decrease the effective height the outer fairing of the vehicle needs to accommodate. The pressure loss in the quick connectors was neglected.



Figure 3.24: Cooling Module Accessories. Left: Quick connectors (couplings). Center: PolyVent plug. Right: Pressure sensor.

3.8.3 Tubing

Polyethylene 13/10 mm soft tubing was used for the coolant loop. Due to its flexibility and ability to be easily cut, assembling the cooling loop is associated with little effort. As an added benefit, soft tubing is easier to integrate into the vehicle, since it can take more intricate paths than hard tubing.

To find the pressure loss over the tubing, large radii of curvature were assumed. With that, the resistance from short corners is negligible. According to Hagen-Poiseuille, the pressure loss over a straight section of tube with length l and radius r is [36]:

$$\Delta p = \dot{V} \frac{8\mu l}{\pi r^4} \quad (3.10)$$

here, the volume flow $\dot{V}$ is 5e-5 m³/s, r = 10mm, and μ the dynamic viscosity of water. With this, the pressure loss for 10 mm internal diameter tubing can be calculated as Δp = 50.9 pa for the tube routing for the whole vehicle. With that, the resistance from the tubing is seen as negligible.

During testing, some leakage was detected on the G1/4" thread between module and quick connectors. Adequate sealing could be achieved by using teflon tape.

3.8.4 Heat Transfer Fluid

For the heat transfer fluid, all simulations and calculations were carried out with pure water. Water, thanks to its high heat capacity and low viscosity makes for good heat transfer and is easy to pump. The initial tests, too, were carried out with tap water. During long-term operation, the water will be replaced with distilled water to protect the electronics in case of leakage and prevent excessive corrosion. Additionally, some glycol might be added if seen fit. Coolant will remain in the water channel during cool-down. Glycol can lower the freezing point of water, which allows for the module to be cooled at lower temperatures without thermal stressing from solidification in the water channel. Ultimately, such lower

temperatures allow for faster cool-down and solidification of the PCM. Additionally, a bactericide and anti-corrosion might be added for long-term application.

3.8.5 Sensors

Three major sensor groups allow for monitoring of the cooling system. All of these will be read over a single sensor bus once the vehicle electronics are functional. The data is saved locally with a high data rate and transmitted at low resolution via WiFi to a Laptop for live view.

As a first sensor group, the MOSFETs all have internal thermistors. These are used for rapidly cut power in case of excessive heating.

Second, two thermistors are placed at the in- and outlet of the module. These consist of standard PT100 thermistors mounted on a G1/4" fitting. At the time of writing, it is unclear whether the precision of these sensors is adequate. The metal fittings are made from brass and heavy, meaning their thermal mass leads to some delay. Additionally, the flow is not necessarily perfectly mixed when passing through the thermistors. As such, it is perfectly possible that the measured temperatures are off by some tenths K. After a long search, it was concluded that thermistors with sensor placement in the flow are not easily available in the needed size.

Third, a single pressure sensor (PBM4-13.B26R.A115.0663 by Baumer [37] as shown in Figure 3.24) is mounted to the module lid. The sensor cable is removable through a threaded plug, which can also be removed using high voltage gloves. The pressure sensor fulfills a safety critical role, as it is used to detect excessive internal pressure. The pressure sensor can be easily removed, revealing a see-through hole for visual inspection of the PCM chamber.

Both the pressure sensor and the module thermistors can be used to calculate the overall liquid fraction of the PCM. If the thermistors are exact enough, an energy balance of the in- and outflowing water can be integrated over time to find the energy absorbed by the module. Additionally, the pressure sensor can measure the thermal expansion the PCM has undergone by measuring the internal pressure. The overall melt fraction can then be computed via the measured expansion. This technique, however, depends on a perfect seal of the modules air/PCM chamber.

During vehicle testing, both these techniques will be implemented on the VCU. If one of two techniques performs reliably, a thermal battery charge indicator will be implemented on the vehicle control software. Otherwise, rapid increases in MOSFET temperatures can be used to detect full melting.

3.9 Safety

In terms of safety, four major hazards were identified.

First, the paraffin is flammable. Candles typically are made from paraffin wax variations. However, the flash point for RT 35 HC is 177 °C [19]. Reaching such temperatures is only regarded as a possibility if other vehicle components, for example the Li-Ion batteries, experience thermal runaway. In a case associated with electrical failure, the paraffin could potentially be ignited since the aluminium module is electrically conductive. However, the internal volume of air is restricted, meaning only a small fraction of the paraffin could burn off. Whether or not this limited burn-off could lead to internal pressures and temperatures which damage the module and lead to a kinetic risk has not been checked.

Thus, under a particular set of circumstances, it cannot be fully guaranteed that the paraffin in the module does not pose a kinetic and thermal risk. However, these circumstances (electrical sparking, thermal runaway of battery, ...) would cause safety risks of much greater degree than the paraffin itself. It can thus be presumed that the risks stemming from the cooling system would be far outweighed by those stemming from the causes of these circumstances.

Second, excessive internal pressure more generally can cause a kinetic risk. While there are few situations that could cause internal pressures over 3 bar, the built-in pressure sensor can be used for monitoring. This data can be viewed in real time thanks to telemetry from the VCU. Based on the data, judgements about safe-to-approach decisions can be made.

Third, there is a risk of the water in the coolant loop damaging electronics. In the worst case, this could lead to a catastrophic short-circuit of power electronics. Thus, some countermeasures were taken to minimize the risk of leakage. All waterblocks are mounted below the MOSFETs. If there is slight leakage, the water will collect at the bottom of the electronics boxes. These boxes are fitted with relief cuts for being bent during manufacturing, where water can escape. Regular visual checks for water, in

particular early on in vehicle testing, will be carried out. Additionally, only distilled water will be used to drastically lower electrical conductivity.

Fourth, RT 35 HC is rated as potentially fatal if ingested or entering airways under GHS08. Thus, during the fill-in procedure and physical contact with the paraffin, protective gear was and will be worn.

3.10 Overview

Table 3.5 shows the overall system mass. The weight limit of 7 kg could be achieved with a safety margin of nearly 400 g. It is likely that this margin will be spent to prevent leakage at the waterblock tubing ports.

Table 3.5: System Mass Overview.

Module	3580 g
Water	450 g
Tubing	230 g
Waterblocks	1720 g
Pump	235 g
Quick Connectors	100 g
Sensors	300 g
Total	6615 g

Figure 3.25 shows the finalized cooling system on the 2022/2023 Swissloop vehicle. The system is also shown schematically in Figure 3.26. Both the resultant volume flow and temperature gradients from heat transfer are discussed below.

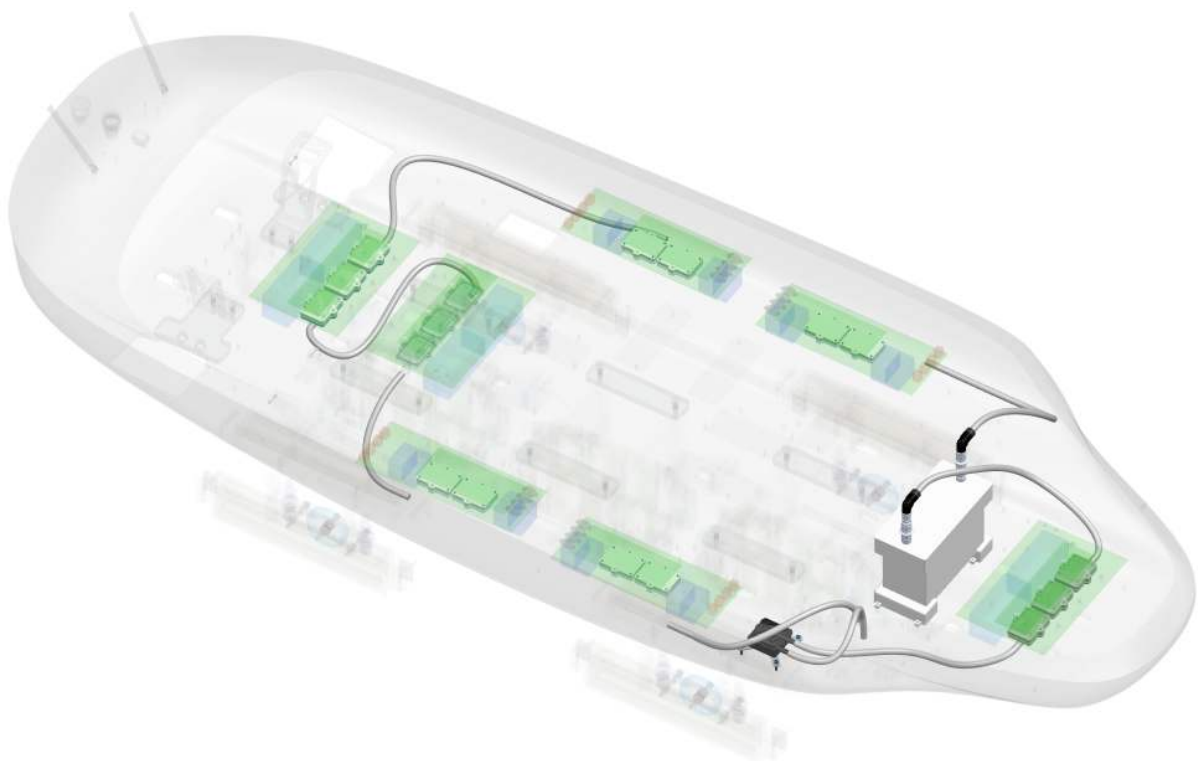


Figure 3.25: System Overview in CAD [Siemens NX].

3.10.1 Pressure Loss and Volume Flow

The pressure loss in the module's water channel has been established in Section 3.6.2. At 180 L/h, Δp_c is 1600 Pa. Similarly, the pressure loss over the waterblocks Δp_{wb} is 1580 Pa at 180 L/h. With 17 waterblocks and one module water channel in series, the overall pressure loss is 28460 Pa. As discussed, the pressure loss over the tubing is negligible and the pump can supply roughly 40 kPa at 180 L/h. Since the pressure generated by the pump, there is ample head room for pressure loss in the couplings, on waterblock tubing ports and elsewhere. A volume flow of at least 180 L/h is expected to be reached.

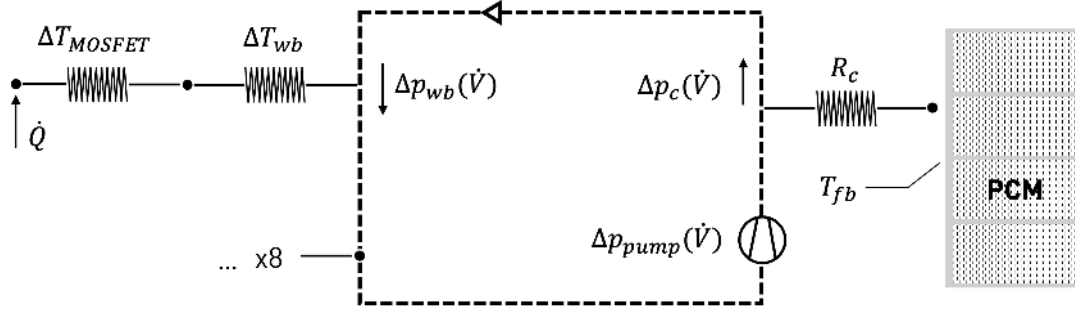


Figure 3.26: System Volume Flow and Heat Transfer Overview. In the water loop, the volume flow results from the pump's head pressure, and the pressure losses in the waterblock and module channel. The MOSFET's internal design as well as the waterblock and channel convective heat transfer introduce thermal resistances into the system.

3.10.2 Heat Transfer

The temperature T_{MOSFET} the MOSFET's chip experiences is given by:

$$T_{MOSFET} = \frac{\dot{Q}}{R_{MOSFET} + R_{wb} + R_c} + T_{fb} \quad (3.11)$$

where R_{wb} is the waterblock resistance, R_c is the convective resistance in the module's water channel, and T_{fb} is the fin base temperature. Here, the delay in the water loop is neglected due to the large transient timeframe of 15 minutes. As before, the levitation scenario is seen as critical for long-term operation. Here, the eight MOSFETs introduce a heat flow of 50 W each into the system. Equation 3.11 can be rewritten as:

$$T_{MOSFET} = \Delta T_{MOSFET} + \Delta T_{wb} + \Delta T_c + T_{fb} \quad (3.12)$$

where ΔT_{wb} and ΔT_c are the temperature gradients at $\dot{v} = 180$ L/h.

The temperature of the fin base T_{fb} is 50 °C. As discussed in Section 3.6.1, this ensures absorption of 440 W over 15 minutes in the PCM. As discussed in Section 3.7, the temperature gradient over the waterblock ΔT_{wb} is 5.42 °C at 150 W during steady state. The linearity of $\Delta T_{wb} = \dot{Q} \cdot R_{wb}$ can be leveraged to adjust ΔT_{wb} to 50 W. Doing so renders $T_{wb} = 1.81$ °C in the levitation scenario. Similarly, the temperature gradient over the module water channel was presented in 3.6.2. ΔT_c equals 3.2 °C at 180 L/h and 440 W. The resistance R_{MOSFET} is 0.328 K/W [26]. For 50 W, ΔT_{MOSFET} is thus 16.4 °C. In total, T_{MOSFET} can be computed using Equation 3.12, and is equal to **71.41 °C**. With this, the MOSFETs are adequately cooled in the critical levitation mode.

Chapter 4

Manufacturing and Testing

4.1 Manufacturing

Most manufacturing steps were carried out by specialized companies. However, some manufacturing processes were not sourced out to allow for multiple design iterations. These include the production of the gaskets and the waterblock lid.

4.1.1 Gaskets

The gaskets used for the waterblocks and module were cut from 1 mm thick rubber mats using a cnc plotter. The plotter, a Zünd M1600 with vacuum table, could be programmed with a .dxf file. The manufacturing of the gaskets caused considerable issues, since the rubber mats were not sufficiently stiff and thus, despite the suction forces of the vacuum table, susceptible to moving when cut. Various cutting strategies were used, with various success. Ultimately, the mediocre cut quality was deemed acceptable for use in preliminary testing. For full-system use, the gaskets will be re-made using a laser cutter.

The waterblock gaskets were made from EPDM Shore A 25. The module gaskets, on the other hand, had to be made from NBR Shore 43, since EPDM is not chemically compatible with paraffin. Particular care was given to avoiding excessive surface roughness on all sealing surfaces. For this reason, the module lid and body were sanded and milled respectively due to the inherent surface qualities from SLS printing. For the waterblock (CNC milled) and its lid (binder jet and SLA printed), it was not deemed necessary to take additional measures relating to surface smoothness.

4.1.2 Waterblock Lid

The waterblock lid posed issues during manufacturing and testing. In order to save weight and limit space usage, it was decided to use a slip-on seal mechanism as shown in Figure 4.1. As an alternative, a G1/4" thread in the lid could have been used to provide an interface for a 90 ° fitting. However, these fitting are only obtainable in brass variants, meaning they are very heavy. For a total of 17 waterblocks, this weight adds up. With a tapped G1/4" thread, the lid could have been made from laser-cut acrylic or similar.



Figure 4.1: Waterblock with Screws and MJF-Printed Lid.

To save weight, the multiple iterations of the lid were 3D-printed using stereolithography (SLA) and multi jet fusion (MJF) processes. During SLA prints, Prusa Tough Orange and Siraya Blu Clear v2 resins were experimented with. For both of these, excellent surface quality on the sealing area of the waterblock could be achieved. However, warping issues were encountered with both resins. Due to time constraints, it was decided to externally manufacture two prototype lids using MJF for reduced warping. Using MJF, the surface roughness increased compared to SLA due to the powder-based manufacturing process. Since the surface quality can drastically impact the performance of the seal, there is a danger that the polyamide MJF lid could increase the likelihood of a leak. For a first prototype, the gasket, base and MJF lid were tested with a pump and no leakage could be detected.

On the slip-on seal at the tubing ports, however, there was significant leakage. This slowed testing down significantly. By using one ear clamps, this issue could be permanently fixed. They allow for the application of very high pressures on the sealing surface and are very lightweight. Thus, they will be used in the full-scale prototype.

A transparent lid can be beneficial during comissioning, since it allows for pockets of air to be detected. Air pockets can cause significant hotspots. However, the waterblock base's design avoids geometries which promote air pockets. Early variants of the SLA-printed lids were transparent, but transparency was later de-prioritized due to warping and leakage issues.

4.1.3 PCM-Fill-In

Filling in the paraffin was done by melting it and simply pouring it into the module. Heating of the paraffin was achieved using an industrial oven at 60 °C and took roughly 8 hours for 12 kg of paraffin in a plastic container. The module lid was specifically designed to enable all fins and paraffin cavities to be visible and easily accessible. Thus, the fill-in procedure was straightforward.

After fill-in, the module was placed in the oven at 60 °C for roughly one hour to eliminate air pockets.

4.2 Validation

At the time of writing, the system has been comissioned. Simple test have been conducted. One waterblock, the module with quick connectors, the DCC pump and a small reservoir could be run in a leakage-free loop over long periods of time.

4.2.1 Initial Testing

In order to demonstrate some degree of functionality, a single MOSFET was cooled. To achieve significant heat outputs, the MOSFET gate was supplied with a constant voltage of roughly 4 V. At a semi-open state, the MOSFET is a very inefficient conductor. With a power supply, a voltage was then applied over the drain/source path of the MOSFET. The resulting current caused internal power dissipation in the MOSFET junction. A multimeter was used to read the resistance values of the MOSFET's internal thermistor. The results are shown in Figure 4.2. Using this method, heat dissipation of up to 180 W was achieved.

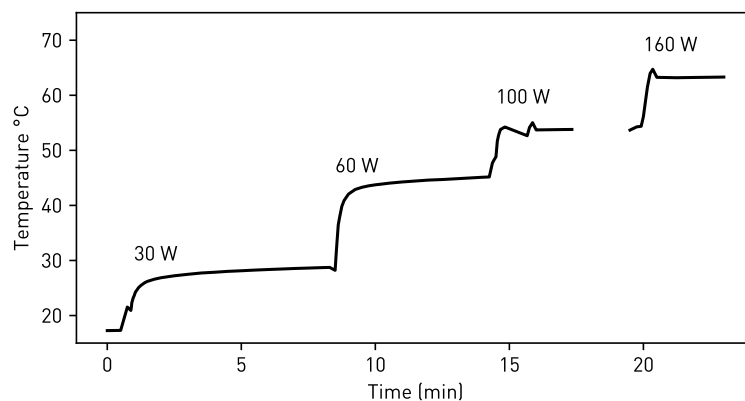


Figure 4.2: First Temperature Measurements upon System Comissioning. Due to the high thermal mass of the water loop and PCM module, the system exhibits quasi-steady-state behaviours.

The waterblock was strapped to the heated MOSFET. Noticeable in Figure 4.2 is how fast the waterblock and MOSFET converge to a quasi-steady-state temperature. This is due to the low thermal mass of the waterblock. Using a single MOSFET at less than 200 W in a cold building, visible PCM melting was not achieved. Most likely, the cold air (17.7 °C) dissipated some of the produced heat on the walls of the module. Additionally, the water in the coolant loop (700 mL) absorbed large amounts of heat. Under load, this amount of water can absorb an additional 49 W when heating by 15 K over 15 minutes.

As shown, the temperature approaches a near steady state level due to the large thermal mass of the complete system. The pump was run at full speed and consumed roughly 15 W. Overall, this test is seen as a success. At 160 W, the MOSFET temperature approached 63.5 °C. With a large internal thermal resistance of 0.328 K/W [26] in the MOSFET, these temperatures were expected to be significantly higher. It is thus assumed that the thermal resistance given by the manufacturer is a very conservative value. Such a discrepancy could be due to the fact that the internal thermistor is on the thermal path between the MOSFET's junction and heat dissipation surface. Thus, the thermistor would experience lower temperatures than the MOSFET itself.

During testing, it was discovered that the couplings introduce slight amounts of air into the system despite the manufacturer claiming the opposite. To account for this, a small reservoir will have to be used. Occasionally, it will have to be vented to let that air escape. After some leakage issues were remedied, a high degree of confidence is put into them after the initial testing.

4.2.2 Operational Testing

Once the 2022/2023 Swissloop vehicle is fully operational, the full system can be tested with large heat inputs. This will, however, occur no earlier than May 2023. As such, the learnings about performance and practical consideration in a functional vehicle can not be discussed here.

4.2.3 Vacuum Testing

If a suitable facility can be found, the module will additionally undergo structural tests in a vacuum chamber. Performing such a test validates the usability of the built system in a vacuum scenario. A high degree of confidence has been obtained from the FEM simulations regarding the structural integrity of the module. For the module gasket, it is near impossible to predict performance at varying pressures.

Chapter 5

Design for Full-Scale

Surrounding the Hyperloop concept, different modes of propulsion and levitation are being developed. In the past decades, many solutions have been proposed with various advantages and drawbacks. Various promising contemporary ideas, like linear switched reluctance motors (LSRMs), have yet to be implemented in a full-scale prototype. No research has been conducted on the high-speed efficiency of LSRMs to the best of the author's knowledge. Partly, this is because most research surrounding LSRMs focuses on low-speed, high thrust designs. Similarly, the levitation systems widely regarded as the most promising among student teams and associated industry partners are (hybrid) electromagnetic suspension designs [38] ((H)EMS). Compared to their electrodynamic suspension (EDS) counterparts, HEMS and EMS systems feature improved drag-to-lift ratios [38]. However, they too have not been built and tested at large speeds yet. Until very recently, assessments of vehicle efficiency have focused on linear inductance motors (LIM) and variants of EDS.

Therefore, a lack of real-world experience regarding vehicle efficiency make the assessment of waste heat nearly impossible. As alluded to, the full-scale vehicle is assumed to require either 4 kW or 8 kW of cooling performance over 1 hour. Important to note, however, is that this represents a very rough estimate. More thorough research of vehicle power usage and thermal management requirements has been omitted due to the breadth of the topic. Instead, focus was put on modelling and manufacturing the small-scale PCM storage solution. As such, a detailed design for a full-scale PCM-based cooling system can not be provided here. Instead, rough weight and capacity estimates are provided here. Additionally, some design considerations for a full-scale system are discussed qualitatively. These include the choice of PCM, heat transfer promotion and system architecture.

5.1 System Architecture

As for the built prototype system, a centralized architecture is favourable in a full-scale scenario. Many of the same advantages apply. The primary of these is fast replacement of liquid PCM. Figure 5.1 shows a concept for automated quick-swapping between vehicle runs. A robotic arm can replace a combined battery and PCM module in a matter of seconds, replenishing electrical energy and thermal capacity. To add to that, a decentralized architecture allows for cooling almost all component types (levitation systems, motor, electronics) thanks to its flexibility. This advantage has been explained in detail in Section 3.2.2.

If longer timeframes for the retrieval of heat can be tolerated, retrieving heat with a coolant loop can provide a good alternative. If a coolant loop is used for flushing, a decentralized architecture can be used. The advantageous heat transfer characteristics can translate to potential weight savings. However, cooling of different component types in various shaped requires significantly more design and simulation work.

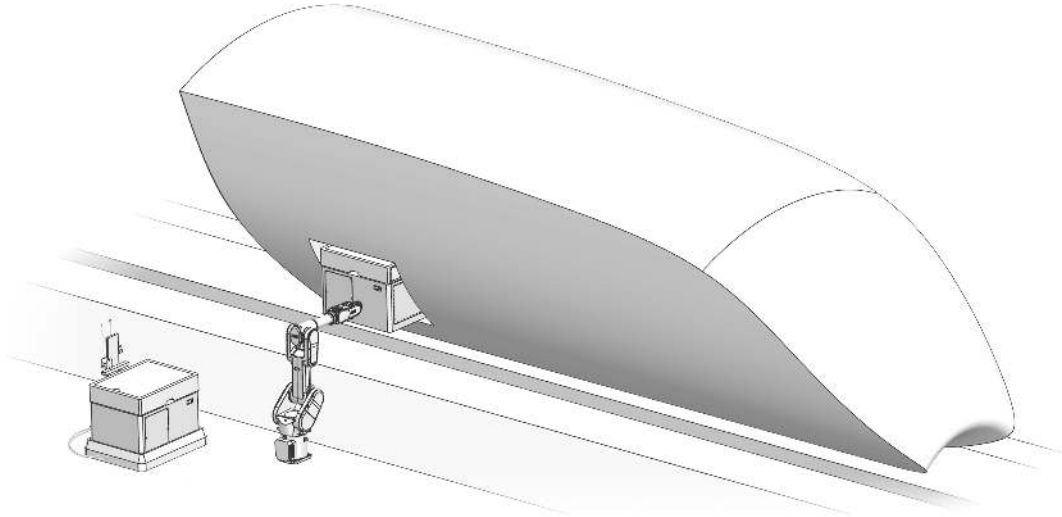


Figure 5.1: Robotic Quick-Swapping Concept for PCM Module. After exchanging the module, it can be cooled over reasonable time frames. The PCM module can be integrated into a single module with a battery for energy storage.

5.2 Choice of PCM

The broad range of phase change mechanisms has been explored in Section 2.4. As discussed, salt hydrates and paraffin are considered the most feasible solutions for full-scale applications. Of these, paraffin was chosen for the small-scale cooling module built here. Due to corrosion, cyclical stability and manufacturer reputation as major downsides salt hydrates were excluded. As such, the decision to deploy paraffin was taken primarily due to practical constraints.

As alluded to in Section 2.4 and shown in Figure 2.6, salt hydrates come with potential weight savings due to their high density and thermal conductivity. The associated decrease in volume and heat transfer requirements promises to reduce enclosure and heatspreader mass. This can potentially offset the disadvantageous mass-specific capacity of salt hydrates. To explore this advantage in detail, large sweeps were conducted in COMSOL to compare three PCM materials.

5.2.1 Simulation

Paraffin, hydrated salt and a paraffin with additives was simulated for a full-scale module. Table 5.1 shows the material values of these. The overall system mass of paraffin with additives compared to hydrated salt was of particular interest. The material data is based on commercially available PCMs [17, 19, 29] and the findings from Section 2.4.4.

Table 5.1: Material Data for Full-Scale PCM Simulations

	Paraffin	Hydrated Salt	Paraffin + 4% CF
h_{pc} (kJ/kg)	210	180	200
k_s (W/mK)	0.2	0.5	1.2
ρ_s (kg/m ³)	880	1570	920
$c_{p,s}$ (kJ/kgK)	2	2	2

Simulations were carried out for four power scenarios. A 440 W, 15 minute load was carried out representing the small-scale module built here. A 4 kW, 15 minute was simulated to represent a low-capacity, high heat transfer scenario. Finally, the full-scale scenarios of 4 kW and 8 kW respectively over 1 hour were simulated. Two parameters, the fin width and amount of fins could be varied. With this variation, the module mass varied. Additionally, the heat transfer shaped by these two parameters was assessed by measuring the liquid fraction.

Model Setup

The model employed here is very similar to that explained earlier in Figure fig:module-comsol-setup. Again, the smallest reoccurring geometric unit was modelled to drastically reduce computational resource intensity. A model representing a centralized module was built as shown in Figure 5.2. To represent a scenario in a flat and narrow vehicle undercarriage, the model depth was set to a fixed value of 1 m, and the module height was set to a constant value of 0.1 m. The wall thickness was set to a constant 5 mm. This is considerably larger than the wall thickness required for the finalized small-scale prototype module. However, with larger module surface area, the internal pressures from expansion and use in vacuum can induce higher stresses. These stresses demand more structural mass. The wall thickness was also kept conservatively high to account for additional accessories, like sensors and tubing ports. This reduced complexity with fixed values was needed to reduce the amount of parameter combinations.

Again, the PCM volume was computed from the needed capacity. As before, the sensible heat capacity of the heatspreader and enclosure was entirely neglected for simplicity. Heating was applied via a constant temperature of 50 °C with $T_m = 35$ °C. Again, this was done to one side of the simulated cell, at the bottom of the module shown here. Convection was neglected. Sufficient heat transfer was assessed through the melt fraction at the final transient time step.

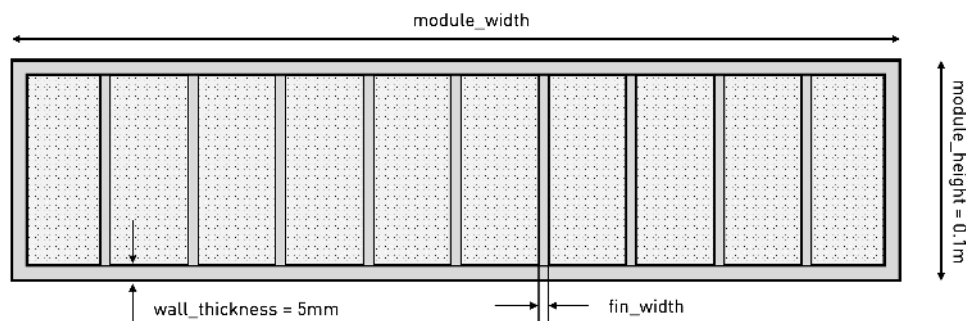


Figure 5.2: Model Setup for Full-Scale PCM Simulations.

Two parameters were varied during the parameter sweeps. The fin width was varied between 1 mm and 10 mm. The number of fins was varied between 1 and 10, or 20 and 120 depending on power scenario. Radial geometries were neglected.

Results

Figure 5.3 shows the results of these sweeps. The melt fraction or liquid fraction LF at the final time step is shown for each parameter combination in color. Mass trade-off lines are also shown. For each material and power scenario, the optimal solution was identified. A necessary condition of achieving 100 % LF was implemented for viable solutions. Of these, the one with minimal overall mass was chosen. The system mass and PCM mass fraction of these optimal solutions are displayed in each plot.

Multiple insights can be formulated from these results. First, system mass obtained for the 440 W, 15 minute scenario is significantly higher than what was achieved in the real world. Whereas the modelled paraffin module weighs at least 8.5 kg in these sweeps, the built module weighs only 3.85 kg as shown earlier. This corresponds to more than a 50 % weight optimization. Mainly, the large difference can be attributed to the assumption of 5 mm outer wall thickness. This assumption is not valid at such low PCM volumes. More generally, this shows that the assumptions made here are very simplistic. Therefore, the mass and PCM fraction of a carefully designed thermal management system could deviate greatly from the values obtained here.

Second, as was the case in earlier results, it can be observed that configurations with a high number of narrow fins are preferable to reduce mass. For example, the optimal geometry for 8 kW over 60 minutes using salt hydrate employs 90 fins of 2 mm width.

Third, the use of additives pays off. In all cases, the simulated full-scale modules which employ additives in paraffin outperform pure paraffin. No further investigation has been conducted, but this is seen as a strong indication that employing graphite or carbon fiber additives comes with significant mass savings depending on the power scenario.

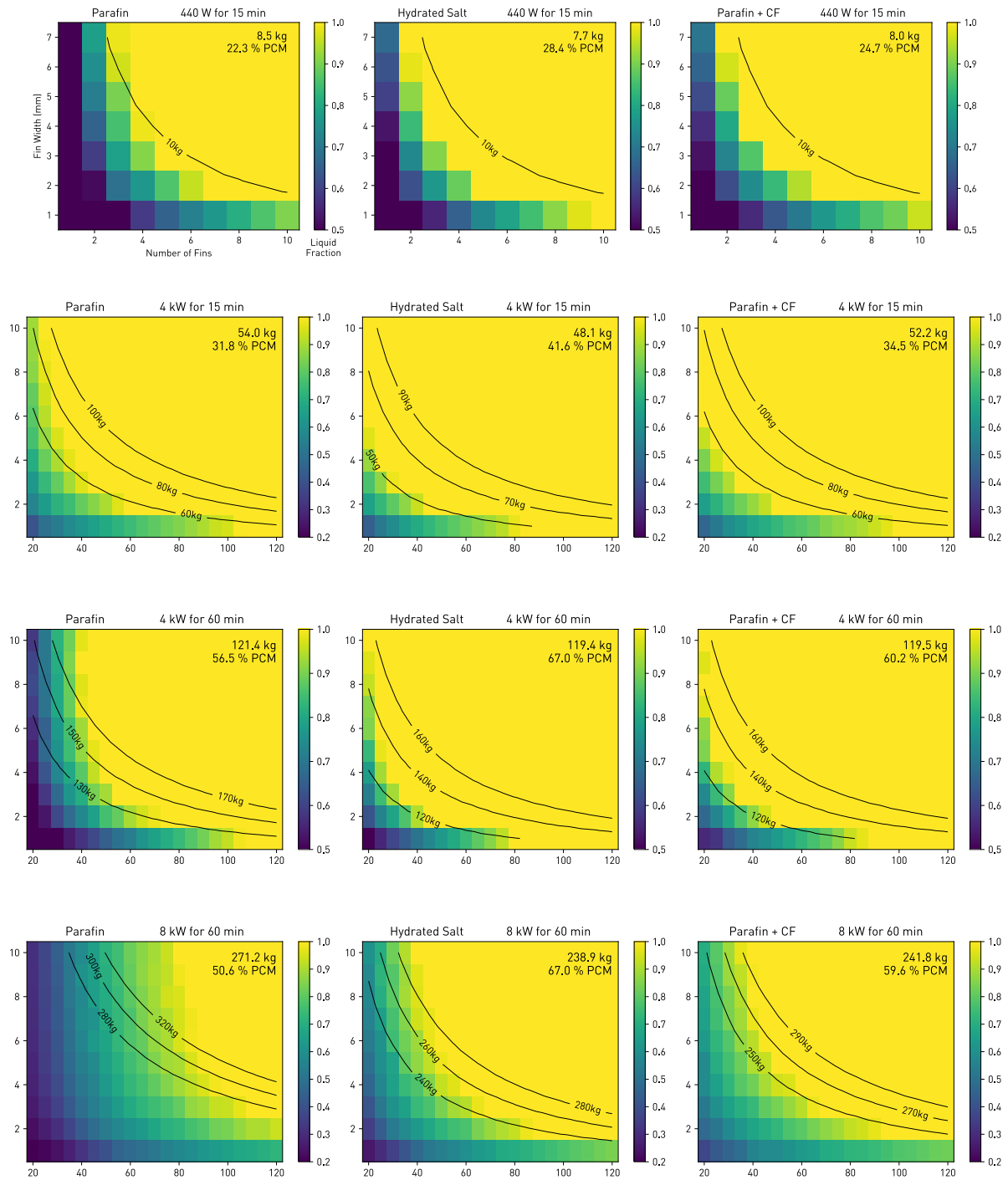


Figure 5.3: Melt Fraction and Overall Module Mass of PCM Types in Full-Scale Scenario. Each data point represents an individually simulated parameter combination.

Fourth, when doubling the capacity, weight roughly doubles despite larger heat transfer. This can, for example, be seen when comparing the 4 kW and 8 kW cases over 1 hour. The effect is explained by the fact that with doubling capacity, PCM volume and fin surface area increase. Thus, provided adequate thermal conductivity in the PCM, the transferred heat can also double. In the optimal solutions, the PCM mass fraction thus stays roughly constant. In both power scenarios, this is the case for both salt hydrates and paraffin with additives.

Lastly, hydrated salt consistently outperforms the other two materials. Despite lower thermal conductivity than paraffin with additives, the mass fraction when using hydrated salt is the highest for all cases. Due to their high density, salt hydrates shorten the effective thermal path through the PCM and thus improve heat transfer. Additionally, the high density gives salt hydrates an advantage in terms of volume occupied by the module. As alluded to earlier, this can be a high priority in full-scale systems, where a passenger cabin is associated with disproportionate space consumption. Overall, hydrated salt is clearly favored over paraffin in a full scale solution from this perspective.

A more obvious observation is that for short, high-power scenarios the PCM mass fraction is low. Since the heatspreader has to be enlarged, it contributed more mass to the overall system. The masses of the optimal solutions from Figure 5.3 are shown in Figure 5.4.

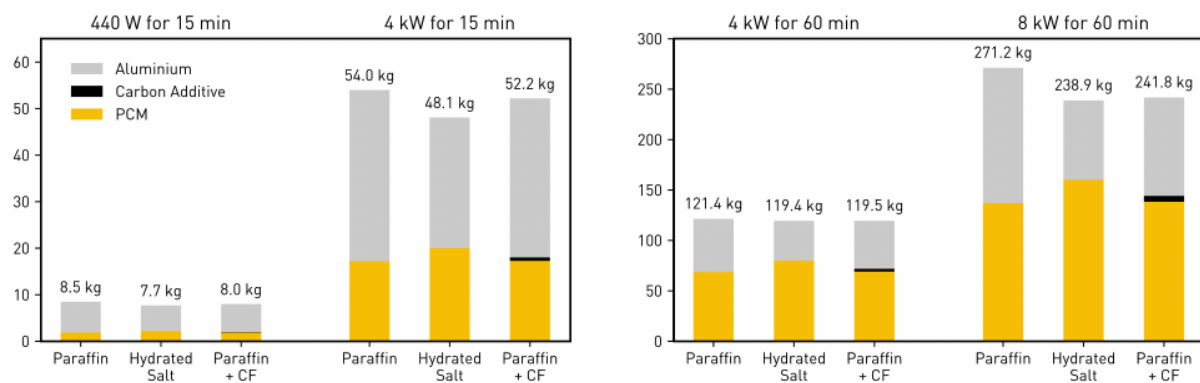


Figure 5.4: Mass Fractions of PCM Types in Full-Scale Scenario.

All of these solutions achieve rather low PCM mass fractions, with a maximum of 67 % PCM for hydrated salt at 8 kW over 60 min. However, the PCM is responsible for the majority of available thermal capacity when warming up by 15 °C. Figure 5.5 illustrates this by showing thermal capacities of paraffin and heatspreader for the optimal solutions.

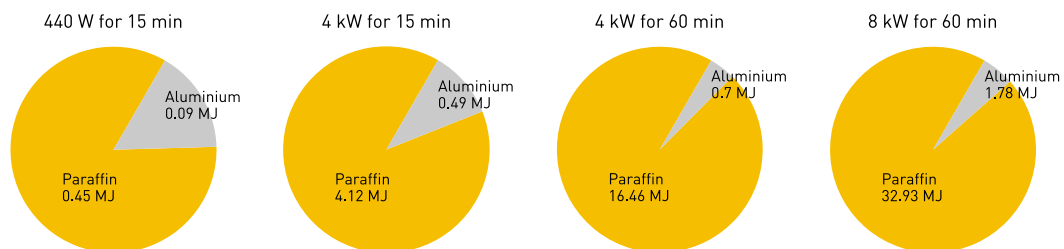


Figure 5.5: Capacity Fraction of Paraffin and Heatspreader in Full-Scale Scenarios. The mass fractions for these configurations can be read from Figure 5.4

Chapter 6

Conclusion

In this thesis, approaches to thermal management in Hyperloop vehicles were discussed. Then, a fully functional prototype of a latent heat storage device was designed, manufactured and commissioned.

Full-Scale Thermal Management Concepts

Various means of thermal management in Hyperloop vehicles were examined to assess their viability. Some concepts can be excluded with a high degree of confidence. Convection at low pressures (10 mbar) has been shown to not provide adequate dissipative performance. Dedicated sensible heat storage, in particular if made from metals, has proved to be significantly heavier than a PCM alternative. Sensible heat storage in a component with combined structural/thermal usage can also be ruled out due to heat leakage and difficult heat retrieval.

Using a liquid-gaseous phase change can be put aside, but not with a high degree of confidence. In terms of mass-specific performance, this approach is very promising. However, retrieval of the vapor from long tube sections poses a great challenge.

Some thermal concepts have been identified as potentially viable, although with some caveats. One of these is the use of water as a sensible heat storage medium. While significantly heavier than its latent PCM counterpart, water is easy to handle and makes for good, direct heat transfer to electronics and other components when pumped. At higher tube pressures of around 100 mbar, convection can potentially be used for thermal management as well. However, adequate airflow over a large and heavy fin array must be ensured. The skin drag such a radiator would introduce is seen as a large downside of this method. It needs to be examined in detail. Radiation, under the tested circumstances, can also provide sufficient dissipation. It can be achieved by leveraging the large surface area of the outer fairing, or by heating a well-isolated probe. For the former, adequate isolation of the fairing to prevent heat leakage is needed. Additionally, ensuring homogeneous fairing temperatures poses a significant challenge, since the fairing's in-plane thermal conductivity needs to be good.

Finally, latent heat storage has been identified as a promising option for full-scale thermal management in Hyperloop applications. As shown, paraffin or salt hydrate PCM units can be built at moderate weight for one hour runs in the single-digit kW range. These units can be scaled easily depending on the power profile. Solid-solid phase change materials were ruled out due to their low enthalpy of phase change.

Phase Change Cooling System

During the design phase of the prototype cooling system, major design considerations such as the choice of components to cool, the architecture, choice of PCM, melting temperature and whether to use additives have been weighed.

Various advantages and drawbacks of centralized and decentralized architecture were discussed. Ultimately, the architectural design of a PCM cooling system is highly dependent on the vehicle type and operational scenario. Here, the choice was made to pursue a centralized approach. By doing so, the module can be quick-swapped, and various component types can easily be integrated into the coolant loop without requiring an adapted PCM module design. Here, water serves as a heat transfer medium, and acts as a buffer during short bursts of heat output.

A melting temperature of roughly 35 °C - 45 °C was determined to be suitable for use on hot summer days. A paraffin-based PCM was chosen due to its large enthalpy of phase change, the abundance of

literature on the material, cyclical stability, availability, manufacturer reputation and non-corrosiveness. A common alternative to paraffin are salt hydrates, which perform worse in these regards and were thus not pursued further. However, the high thermal conductivity and high density of salt hydrates can enhance heat transfer greatly. This leads to a decrease in heatspreader mass. Additionally, the low volume makes for more compact systems.

The use of additives to enhance thermal conductivity was omitted due to time constraints. However, as has been shown in Section 5.2.1, additives can increase thermal conductivity in the PCM drastically and thus reduce heatspreader mass. There is extensive literature on the subject, which can be used in future work.

For the heatspreader, a central channel approach with thin fins was used here. While the performance of this approach has been found to be adequate for the application at hand, other geometries could potentially be investigated in the future.

Two major challenges which were not expected at the outset were encountered while designing the prototype PCM module. First, thermal expansion poses a large problem for closed PCM containers. Structural reinforcement and enlarged air pockets to buffer the effect are indispensable. Second, PCMs can affect the choice of materials in the containers used. For example, a gasket material (NBR) which can tolerate paraffin had to be chosen.

Some design features, like vacuum-proofing and hot-weather resilience, have yet to be tested. The system's reliability will also only be assessed after several months of use in the Swissloop vehicle.

Chapter 7

Outlook

After this thesis has concluded, a second module will be manufactured using the SLS process. This allows for quick-swapping during operation. Additionally, a larger number of waterblock lids and gaskets will be made. The complete prototype vehicle cooling system will be assembled and used during testing of the 2022/2023 Swissloop vehicle. In July 2023, the cooling system will be demonstrated along with the vehicle at European Hyperloop week. The learnings regarding reliability, everyday usage and real-world thermal performance will be used for planning a second generation PCM cooling system for Swissloop vehicles.

Regarding further examination of topics on thermal management in near-vacuum tube environments, multiple topics could be pursued:

1. Using a vehicle's outer fairing for radiative dissipation poses issues which could be tackled. Specifically, heat transfer from electronics to the fairing and along the fairing's surface could be investigated in detail.
2. A system employing a heat pump to warm up a small radiation probe could be built for a small- or full-scale system.
3. Drag and convective heat dissipation of a fin array at 50 - 100 mbar could be investigated using CFD methods.
4. A study further investigating optimal additive ratios and the use of salt hydrates could be carried out. From a modelling perspective, many open questions can still be answered regarding trade-offs between the thermal conductivity and capacity of PCMs.
5. Some of the drawbacks which prevented the use of salt hydrates in this thesis could be examined in detail. If these drawbacks (corrosion, cyclical stability, manufacturer reputation and availability) can be addressed, salt hydrates are an attractive alternative to paraffin. Their thermal conductivity and density makes for improved heat transfer, resulting in weight savings in the heat spreader. Additionally, more compact cooling modules can be built using salt hydrates.
6. An autonomous system for rapid quick-swapping of PCM modules could be designed.
7. The cooling of Li-Ion battery cells using PCMs could be examined in detail. The battery as a heat source has been entirely neglected for the purposes of this thesis, but might indeed be relevant in full-scale use. Of additional interest is the design of a combined PCM heat storage and electrical battery module.

Bibliography

- [1] Swissloop. *Swissloop*. 2023. URL: <https://swissloop.ch/>.
- [2] Alois Feusi. *Simona de Silvestro auf Testfahrt in Duebendorf*. Zurich, Sept. 2020. URL: <https://www.nzz.ch/zuerich/simona-de-silvestro-auf-testfahrt-in-duebendorf-ld.1577461>.
- [3] Denis Tudor and Mario Paolone. "Operational-driven optimal-design of a hyperloop system". In: *Transportation Engineering* 5 (Sept. 2021), p. 100079. ISSN: 2666-691X. DOI: 10.1016/J.TRENG.2021.100079.
- [4] Nishanth Dongari, Ashutosh Sharma IITK, and F Durst. "Pressure-driven diffusive gas flows in micro-channels: From the Knudsen to the continuum regimes". In: *Microfluidics and Nanofluidics* 6 (Jan. 2009), pp. 679–692. DOI: 10.1007/s10404-008-0344-y.
- [5] VDI Gesellschaft Verfahrenstechnik und Chemieingenieurwesen. *VDI-Wärmeatlas, 11. Auflage*. Tech. rep. 2013.
- [6] Paul Allen Tipler and Gene Mosca. *Physik für Wissenschaftler und Ingenieure*. 7th ed. Berlin; Heidelberg: Springer Spektrum, 2015, pp. 556–558.
- [7] Laboratory of Thermodynamics in Emerging Technologies. *Formelsammlung Thermodynamik II – Teil II*. Zurich: ETH Zurich, 2021.
- [8] H. Wu et al. "Characterization of thermal cross-talk in a MEMS-based thermopile detector array". In: *Journal of Micromechanics and Microengineering* 19.7 (2009). ISSN: 09601317. DOI: 10.1088/0960-1317/19/7/074022.
- [9] Federico Lluesma et al. "Evacuated-Tube, High-Speed, Autonomous Maglev (Hyperloop) Transport System for Long-Distance Travel: An overview". In: *IEEE Electrification Magazine* 9.4 (Dec. 2021), pp. 67–73. ISSN: 23255889. DOI: 10.1109/MELE.2021.3115543.
- [10] Reza Shokrollah Abhari and Aldo Steinfeld. *List of Equations for Thermodynamics III*. PREC, 2021.
- [11] Moran and Shapiro. *Fundamentals of Engineering Thermodynamics: Appendix*. 3rd ed. New York: J. Wiley & Sons, 1995. ISBN: 978-0471076810.
- [12] National Institute of Standards and Technology. *NIST Chemistry WebBook, SRD 69*. 2021. URL: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C75150&Mask=4>.
- [13] National Institute of Standards and Technology. *NIST Chemistry WebBook, SRD 69*. 2021. URL: <https://webbook.nist.gov/cgi/inchi?ID=C79209&Units=SI&Mask=4#Thermo-Phase>.
- [14] Ali Fallahi et al. *Review on solid-solid phase change materials for thermal energy storage: Molecular structure and thermal properties*. 2017. DOI: 10.1016/j.applthermaleng.2017.08.161.
- [15] PCM Products Ltd. *PlusICE Solid-Solid (X) Range*. 2018. URL: <https://www.pcmproducts.net/files/X%20range-2018.pdf>.
- [16] Lingai Luo and Nolwenn Le Pierrès. "Innovative Systems for Storage of Thermal Solar Energy in Buildings". In: *Solar Energy Storage* (Jan. 2015), pp. 27–62. DOI: 10.1016/B978-0-12-409540-3.00003-7.
- [17] Rubitherm Technologies GmbH. *Technical Datasheet SP 31*. 2022. URL: https://www.rubitherm.eu/media/products/datasheets/Techdata_-SP31_DE_12072022.PDF.
- [18] Rubitherm Technologies GmbH. *Technical Datasheet SP 70*. 2020. URL: https://www.rubitherm.eu/media/products/datasheets/Techdata_-SP70_DE_24112020.PDF.
- [19] Rubitherm Technologies GmbH. *Technical Datasheet RT35HC*. 2020. URL: https://www.rubitherm.eu/media/products/datasheets/Techdata_-RT35HC_EN_09102020.PDF.

- [20] F. Frusteri et al. "Thermal conductivity measurement of a PCM based storage system containing carbon fibers". In: *Applied Thermal Engineering* 25.11-12 (Aug. 2005), pp. 1623–1633. ISSN: 13594311. DOI: 10.1016/j.applthermaleng.2004.10.007.
- [21] Jun Fukai et al. *Thermal conductivity enhancement of energy storage media using carbon Fibers*. Tech. rep. URL: www.elsevier.com/locate/enconman.
- [22] Ali Karaipekli, Ahmet Sari, and Kamil Kaygusuz. "Thermal conductivity improvement of stearic acid using expanded graphite and carbon fiber for energy storage applications". In: *Renewable Energy* 32.13 (Oct. 2007), pp. 2201–2210. ISSN: 09601481. DOI: 10.1016/j.renene.2006.11.011.
- [23] V. Shatikian, G. Ziskind, and R. Letan. "Numerical investigation of a PCM-based heat sink with internal fins". In: *International Journal of Heat and Mass Transfer* 48.17 (Aug. 2005), pp. 3689–3706. ISSN: 00179310. DOI: 10.1016/j.ijheatmasstransfer.2004.10.042. URL: shatikian05.
- [24] Ricco Eberhard. *Swissloop Focus Project 20/21 Final Report*. Tech. rep. Swissloop, 2021.
- [25] Luvata (Mitsubishi Materials Corp.) *Copper Hollow Conductors*. Tech. rep. 2018. URL: https://cdn.luvata.com/docs/default-source/products/hollow-conductors/copper-hollow-conductors_eng.pdf?sfvrsn=d5ed800a_25.
- [26] Infineon Technologies AG. *Datasheet FF6MR12W2M1_B11*. München, Germany, 2019.
- [27] Carl Brander and Lukas von Briel. "Swissloop Focus Project 21/22 Final Report". In: (2022).
- [28] Infineon Technologies AG. "Datasheet F4-33MR12W1M1H_B76". In: (2022).
- [29] Rubitherm Technologies GmbH. *Rubitherm Webpage PCM RT-Serie*. 2023. URL: <https://www.rubitherm.eu/produktkategorie/organische-pcm-rt>.
- [30] Ali C Kheirabadi and Dominic Groulx. *SIMULATING PHASE CHANGE HEAT TRANSFER USING COMSOL AND FLUENT: EFFECT OF THE MUSHY-ZONE CONSTANT*. Tech. rep. 6. 2015, pp. 427–440.
- [31] GE Additive. *Datasheet AlSi10Mg M2 Multilaser*.
- [32] W. L. Gore & Associates GmbH. *GORE® Protective Vents Catalogue*. 2021. URL: https://www.distrelec.ch/Web/Downloads/_t/ds/WLGore_ScrewIn_Vents_eng_tds.pdf.
- [33] Raphaël N. Glaesener et al. "Continuum representation of nonlinear three-dimensional periodic truss networks by on-the-fly homogenization". In: *International Journal of Solids and Structures* 206 (Dec. 2020), pp. 101–113. ISSN: 00207683. DOI: 10.1016/j.ijsolstr.2020.08.013.
- [34] V. S. Deshpande, N. A. Fleck, and M. F. Ashby. "Effective properties of the octet-truss lattice material". In: *Journal of the Mechanics and Physics of Solids* 49.8 (Aug. 2001), pp. 1747–1769. ISSN: 0022-5096. DOI: 10.1016/S0022-5096(01)00010-2.
- [35] HABA AG. *Datasheet HABA G-Alu25*. 2022. URL: https://www.haba.ch/fileadmin/user_upload/Produktblaetter_deu/Alu_G-Alu25_DE.pdf.
- [36] P. Jenny and T. Rösgen. *Fluidodynamik I/II - Skript zur Vorlesung*. Zurich: Institut für Fluidodynamik, 2021, pp. 52–53.
- [37] Baumer Electric AG. *Datasheet Baumer PBM4*. Frauenfeld, 2021. URL: https://www.baumer.com/medias/__secure__/Baumer_PBM4_EN_20210614_DS.pdf?mediaPK=9022666702878.
- [38] Jonas Kristiansen Noland. "Prospects and Challenges of the Hyperloop Transportation System: A Systematic Technology Review". In: *IEEE Access* 9 (2021), pp. 28439–28458. ISSN: 21693536. DOI: 10.1109/ACCESS.2021.3057788.